

SC-550: A Sintered Calcite Reference Material for *In Situ* U-Pb Geochronology

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The accuracy and precision of *in situ* U-Pb geochronology of carbonates is limited by the intrinsic isotopic heterogeneity of natural reference materials (RMs). Recent efforts aimed at producing homogenous carbonate RMs through nano-milling of natural samples or synthesis of amorphous calcium carbonate have shown promising results. Here we present a new, microcrystalline calcite material – SC-550 (Sintered Calcite 550 °C) – with element mass fractions, Sr and U/Pb isotopic ratios and ablation behaviour comparable to that of a natural carbonate. A microbial limestone is first milled to nano-sized particles and then sintered at high pressure and temperature using an internally heated pressure vessel (IHPV) apparatus. Sample heterogeneity is removed through milling, while textural coarsening is aimed at generating ablation behaviour like that of natural calcite materials. Ideal synthesis conditions were achieved at 500 MPa, 550 °C, and 24 h, where the material is texturally and compositionally homogeneous at the microscale (tens of μm). Inter- and intra-experimental element mass fraction homogeneity is determined using EPMA (Ca mass fraction) and LA-ICP-MS analyses in two laboratories using static spot analyses and area mapping. The results for elements relevant for environmental and palaeoclimate applications show relative standard deviation (RSD) values in the range of 5–10%. The $^{87}\text{Sr}/^{86}\text{Sr}$ ratio of the starting nanoparticulate powder is determined by solution MC-ICP-MS as 0.708104 ± 0.000002 (2s) and the Sr isotopic micro-homogeneity in experimental charges is confirmed by LA-MC-ICP-MS. The U–Pb isotopic ratios of the nanoparticulate powder determined by ID-TIMS are $^{207}\text{Pb}/^{206}\text{Pb} = 0.66226 \pm 0.00012$ (2s) and $^{238}\text{U}/^{206}\text{Pb} = 4.6244 \pm 0.0057$ (2s). Tests of U/Pb ratio homogeneity in experimental charges by LA-ICP-MS show < 0.6% variation (RSD) for $^{207}\text{Pb}/^{206}\text{Pb}$ and < 3% variation for $^{238}\text{U}/^{206}\text{Pb}$. The suitability of the sintered calcite material as a primary RM for U-Pb geochronology of carbonates is confirmed by testing ablation and laser induced fractionation behaviour compared with natural carbonates and reproducing published dates within uncertainty limit for a set of natural calcite materials.

Keywords: laser ablation-inductively coupled plasma-mass spectrometry, calcium carbonate, nanoparticulate powder, IHPV, reference material, strontium isotopes.

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Calcite and other carbonate minerals are fundamental components of many terrestrial environments, from the

hydrosphere (i.e., micro- and macro-organisms' shells in marine environments) to the lithosphere (e.g., palaeosols, speleothems, fracture and fault mineralisation) and the biosphere (e.g., fossilised tissues). The precipitation of calcite throughout Earth's history (i.e., at least since 3.4 Ga;

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Lowe 1980, Byerly *et al.* 1986) makes this mineral an unparalleled archive, holding key information on the long-term evolution of the crust, climate, and life. This information is preserved in calcite elemental and isotopic composition; however, the timing of specific geological or biological processes can only be assessed with knowledge of the absolute age of formation. Direct dating of calcite taking advantage of the U-Pb decay chains, although successfully developed starting from the 1980s (Jahn and Cuvellier 1994, Rasbury and Cole 2009), has only recently seen a steep rise in applications due to advances in laser ablation-inductively coupled plasma-mass spectrometry (LA-ICP-MS) (Roberts *et al.* 2020). The relative ease of use, high sample throughput and spatial resolution provided by this technique have allowed a broad range of applications, which span from sedimentary and diagenetic processes (e.g., Methner *et al.* 2016, Godeau *et al.* 2018, Montano *et al.* 2021, Looser *et al.* 2026), to tectonic evolution (Roberts and Walker 2016, Nuriel *et al.* 2017, 2019, Hansman *et al.* 2018, Looser *et al.* 2021), ore deposit formation (e.g., Burisch *et al.* 2017, Giorno *et al.* 2022, Mottram *et al.* 2024, Tavazzani *et al.* 2024, Bos Orent *et al.* 2025), palaeoclimate reconstruction (e.g., Drake *et al.* 2017, Aguirre Palafox *et al.* 2024, Markowska *et al.* 2025) and palaeontological investigation (e.g., Li *et al.* 2014, Scardia *et al.* 2019, Qi *et al.* 2024).

An intrinsic limitation of LA-ICP-MS is that *in situ* analyses rely on matrix-matched primary reference materials (RMs) whose ablation, transport and ionisation characteristics should ideally result in fractionating elements and isotopes in the same way as in the unknown sample of interest (Sylvester 2008, Hathorne *et al.* 2008, Jochum *et al.* 2012, 2014). Carbonate RMs currently in use (a recent overview can be found in Li *et al.* 2026), even if homogeneous in age, are heterogeneous in U/Pb isotopic ratios and initial Pb concentration. Consequently, rather than correcting signals from individual analytical spots as is customary for applications where RMs are homogeneous in U/Pb ratios (e.g., zircon geochronology), raw results need to be corrected for elemental and isotopic fractionation by a session-specific factor that reproduces the reference isochron age of the RM. This introduces a lengthy data-reduction procedure and adds uncertainty related to age homogeneity of the chosen RM (e.g., 1.5–2.5%, Roberts *et al.* 2017, Guillong *et al.* 2020, 2024). The microanalytical community would thus benefit from a carbonate RM homogenous in Pb and U/Pb isotopic ratios for wide distribution. Ideally, such a RM should also match the ablation behaviour of the unknown carbonate and consequently all “matrix effects” (Jochum *et al.* 2014, Guillong *et al.* 2020), including the amount and depth progression of inter-element laser induced elemental

fractionation (LIEF). That way, raw results can be corrected and uncertainties propagated to the final U-Pb age of an unknown sample following commonly accepted community derived criteria (e.g., Horstwood *et al.* 2016).

Recent years have seen substantial progress in the production of synthetic (*sensu lato*) materials whose elemental and isotopic homogeneity at the micro- to nano-scale allows them to be used as matrix-matched RMs for different analytical applications, including LA-ICP-MS. This extends to the production of custom-made carbonate materials, which have been realised by a suite of different techniques:

- (i) Milling of natural calcite to sub-micrometre grain sizes (i.e., < 1.5 μm , Garbe-Schönberg and Müller 2014) to produce pressed powder pellets with sufficient elemental and isotopic homogeneity to be used as microanalytical RMs for trace element and Sr isotopic measurements (MACS-3NP, JCp-1NP and JCI-1NP, Weber *et al.* 2018, Jochum *et al.* 2019, NFHS-2-NP, Boer *et al.* 2022).
- (ii) The flame spray technique (Tabersky *et al.* 2014) to produce matrix- and composition-customised materials from a liquid organic precursor that incorporates elements of interest (e.g., Sr, REE; NanoSr, Weber *et al.* 2020).
- (iii) Synthesis of calcite with known elemental and isotopic composition by crystallisation from doped amorphous calcium carbonate (ACC, Addadi *et al.* 2003) at room temperature (Meece and Benninger 1993, Reeder *et al.* 2000, 2001) or through heating and pressurisation (Koga and Nakagoe 1998, Yoshino *et al.* 2012).

Regarding (iii), recently, Miyajima *et al.* (2021) geared this technique to produce a material homogeneous in U and Pb mass fraction and U/Pb isotopic ratios at the scale of LA-ICP-MS analyses (tens of μm). However, unmitigated issues with U uptake from the dopant solution and with Pb contamination during ACC preparation caused large variability in U/Pb ratios of the final synthetic products. At present this nanomaterial is not suitable as RM for normalisation of U/Pb ratios (Miyajima *et al.* 2021).

Here we present a new protocol for the production of a microcrystalline calcite material through sintering of a nano-milled natural carbonate sample. This protocol takes advantage of the well-established nano-milling process developed by Garbe-Schönberg and Müller (2014) to obtain a calcite material with a greater compositional and particle size homogeneity compared with regularly milled

powders (Jochum *et al.* 2019). In contrast to existing RMs (e.g., Jochum *et al.* 2019, Boer *et al.* 2022), the nanoparticulate powder is not cold-pressed into a pellet but sintered at high temperature ($> 500\text{ }^{\circ}\text{C}$). At these conditions, grain growth, volume diffusion and grain-boundary diffusion lead to material densification without decomposition of the starting nanoparticulate powder (Olgaard and Evans 1988, Covey-Crump 1997, Davis *et al.* 2011). This combined approach ensures that sample heterogeneity is removed through milling, while densification and textural coarsening are aimed at generating ablation behaviour similar to typical natural calcite samples. Microanalytical testing demonstrates that a set of experimental charges is suitable as a matrix-matched calcite RM (SC-550, Sintered Calcite 550 $^{\circ}\text{C}$) for *in situ* measurements of U/Pb isotopic ratios. Moreover, SC-550 demonstrates a good compositional homogeneity in a suite of elements used across geochemical applications (e.g., Mg, Mn, Fe, Ba, Sr, REE) and in Sr isotopic composition, making it a potential candidate as primary or quality control RM for *in situ* measurements of Sr isotopic ratios and elemental mass fractions in calcite samples.

Material and methods

Starting material

The starting material for our sintering experiments was a batch of powder produced from the Carboniferous microbial boundstone lithoclast RA138, previously characterised by Guillong *et al.* (2024). RA138 contains four texturally and compositionally distinct carbonate generations (Figure S1). Of the four phases, the high U ($\sim 4\text{ }\mu\text{g g}^{-1}$) and U/Pb, botryoidal cement C1 is characterised by a good level of age homogeneity, making it a suitable natural RM for carbonate U-Pb geochronology (Guillong *et al.* 2024). As our study required large amounts of starting material, we did not separate a specific carbonate generation but rather used the bulk sample including all four identified cement types. As such, its U/Pb ratio is lower than that of pure C1 cement, but it is comparable to U/Pb in many natural carbonates (see Roberts *et al.* 2020 for a compilation). A large batch of nanoparticulate powder (240 g) was obtained by nano-milling conducted by myStandards GmbH (Kiel, Germany). The milling procedure, which includes freeze-drying and homogenisation, follows the protocol described in Garbe-Schönberg and Müller (2014) and produces material with nominal grain size $d_{50} < 1.5\text{ }\mu\text{m}$.

High-pressure experiments

We first conducted five preliminary, small volume sintering experiments using: (i) a piston-cylinder apparatus (PCA) and (ii) an externally heated pressure vessel (EHPV, also known as “cold seal”), the set up of which and results are summarised in Appendix S1 and Appendix S9. These experiments served to explore suitable pressure and temperature conditions of sintering, but the experimental capsules in these devices are too small to produce enough material to be widely distributed ($\sim 0.2\text{ cm}^3$). For the goal of producing a larger volume of characterised RM, we performed six internally heated pressure vessel (IHPV) experiments where capsules can be much larger ($\sim 2.5\text{ cm}^3$, Table 1).

Individual IHPV experiments used ca. 2.2 g of nanoparticulate powder packed into $9.5 \times 50\text{ mm}$ Ag capsules (Figure 1a). Experimental runs were performed for 4 and 24 h in a vertically-oriented internally heated pressure vessel (IHPV) at the Leibniz University Hannover (Berndt *et al.* 2002) at 550, 650, and 800 $^{\circ}\text{C}$ and 500 MPa (Table 1). Argon was used as a pressure medium, and temperature was controlled and monitored using three K-type (NiCr-NiAl) thermocouples to establish a ca. 5 cm wide thermal hot zone with a temperature gradient below 15 $^{\circ}\text{C}$. Pressure was monitored with a Burster pressure transducer and automatically controlled at the target value with a maximum offset of 5 MPa. Experiments were directly heated to target temperatures with a 50 $^{\circ}\text{C min}^{-1}$ heating rate. To avoid overshooting the temperature, heating rates were reduced to 10 $^{\circ}\text{C min}^{-1}$ in proximity (30–50 $^{\circ}\text{C}$) of the final equilibration temperature, and temperature fluctuations at run conditions were generally below 1 $^{\circ}\text{C}$. Quenching was performed by switching off the furnace’s power supply, resulting in initial quenching rates of ca. 200 $^{\circ}\text{C min}^{-1}$. Before the experiments, capsules were tested for tightness by immersion in acetone while monitoring potential mass changes. Capsules exhibiting any weight variation beyond the uncertainty of the microbalance were discarded. Similarly, after the experiments, recovered capsules were re-weighed and only runs showing no weight changes were considered successful.

Analytical techniques

To assess inter-experiment reproducibility and intra-capsule homogeneity, the IHPV capsules were cut to obtain three 2 mm-thick wafers, one from the middle and one from each termination of the capsule (Figure 1b, c, d). All wafers were fixed in 1” Specifix resin mounts using a Struers ApS (Denmark) mounting press, followed by grinding and mechanical polishing to 1 μm and finished with a

Table 1.
Experimental conditions and capsule information

Run	Date (dd.mm.yy)	Experimental conditions			Capsule property			
		P (MPa)	T (°C)	t (h)	Diameter (mm)	Height (mm)	Mass (mg)	Capsule material
SC-A	03.05.24	500	800	24	9.5	50	1962	Ag
SC-B	26.06.24	500	800	4	9.5	50	1970	Ag
SC-G	20.12.24	500	650	24	9.5	50	2263	Ag
SC-C	03.07.24	500	550	24	9.5	50	2228	Ag
SC-D	01.08.24	500	550	24	9.5	50	2188	Ag
SC-E	02.08.24	500	550	24	9.5	50	2122	Ag

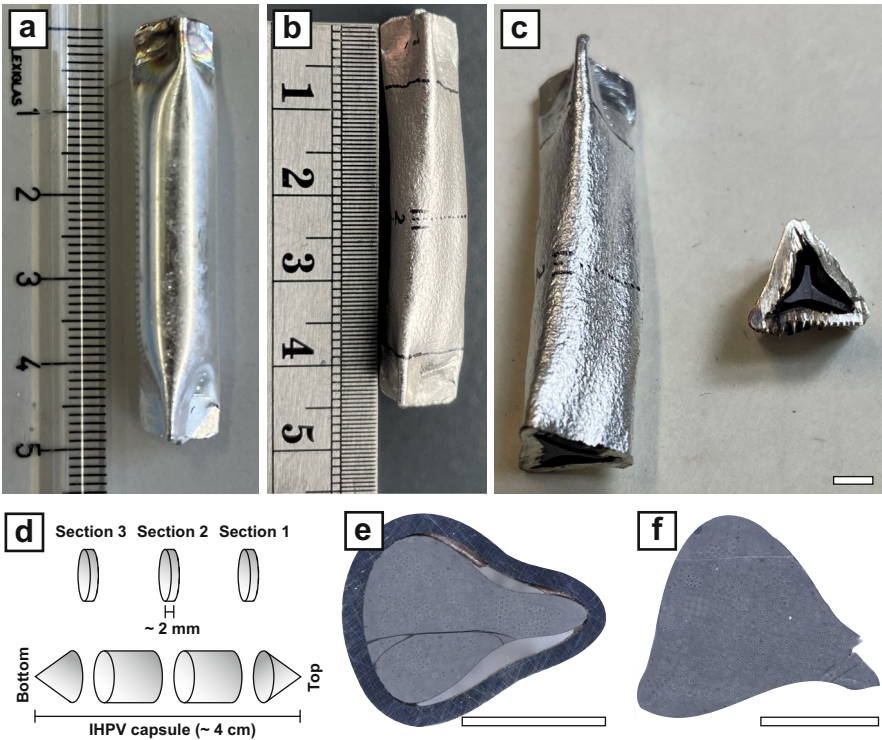


Figure 1. Capsule and macro aspect of Internally Heated Pressure Vessel (IHPV) experiments. View of silver capsule containing carbonate nanoparticulate powder (a) before and (b) after an IHPV run. (c) Frontal view of capsule after microdiamond wire saw slicing of lowermost portion. (d) Schematic of experimental capsule sampling to test compositional homogeneity across the length of the capsule. (e, f) Top view, optical microscope image of slices from the central section of two experimental charges with cross sectional length of ca. 9.5 mm and thickness of 2 mm. LA-ICP-MS ablation pits are visible. Scale bar is 5 mm across all images.

colloidal-silica suspension to ensure a damage-free surface for electron backscatter diffraction (EBSD) (Figure 1.e, f).

Additionally, two grams of nanoparticulate powder were loaded in a cylindrical tungsten carbide mould and were pressed at room temperature with 200 bar hydraulic pressure on a tungsten carbide piston for 10 min. This produced one powder pellet, 10 mm in diameter and 2 mm thick.

The pressed powder pellet and the experimental charges were characterised using several imaging and analytical techniques, outlined below.

Imaging

Secondary electron (SE) imaging, backscatter electron (BSE) imaging, energy-dispersive x-ray spectroscopy (EDS)

and EBSD data were obtained with a TESCAN Solaris field emission gun scanning electron microscope (FEG-SEM), equipped with an Oxford Instruments Symmetry EBSD detector at the Department of Geosciences, University of Padova. BSE and SE imaging conditions were 15 kV acceleration voltage, 3 nA beam current, and 5 mm working distance for large overview images and 5 kV acceleration voltage, 300 pA beam current, and 4 mm working distance for high-resolution images. EBSD maps for measurements of clast size were indexed and filtered with Oxford Instruments AZtec software and further processed with MTEX toolbox version 5.6.1 in MATLAB (Hielscher and Schaeben 2008). Acquisition conditions for EBSD maps were 25 kV of acceleration voltage, 5 nA of current, 14 mm of working distance, and a step size of 0.15 μm . For pore size analysis, BSE images were manually thresholded and measured using the image analysis software ImageJ. Four EBSD maps (i.e., experimental charges SC-A, SC-D, SC-E and SC-G) were acquired for grain size analysis and were processed using MTEX.

Geometry of LA-ICP-MS craters was characterised at ETH Zürich with a VHX-7000 digital microscope (Keyence Corporation, Japan) equipped with 3D image capture and depth measurement capabilities. The ablation rate (nm per pulse) was calculated by division of crater depth by number of pulses.

Element mass fractions

Element mass fractions were determined by *in situ* techniques to test for inter- and intra-capsule homogeneity.

EPMA: Two embedded sintered carbonate slices (SC-B, 800 °C and SC-E, 550 °C) were polished, cleaned and carbon coated (~ 20 nm). Analyses were carried out at ETH Zürich using a JEOL JXA-8230 electron probe microanalyser (EPMA). Additional details are included in Appendix S1.

LA-ICP-MS: Trace element measurements (static spots) were performed at ETH Zürich using an ASI (now Applied Spectra) RESolution-LR 193 nm excimer laser ablation system coupled to a Thermo Element XR single-collector sector field ICP-MS following a similar workflow of Guillong *et al.* (2024) and Tavazzani *et al.* (2024). Detailed description of the methodology is included in Appendix S1.

Trace element distribution within three selected experimental charges was mapped by LA-ICP-MS at the Natural History Museum in London using a Teledyne Iridia LA system equipped with a Cobalt fast-washout ablation cell coupled

to an Agilent 8900 ICP-MS equipped with a quad-lock system (Norris *et al.* 2021). The following signals were monitored during the session: ^{25}Mg , ^{43}Ca , ^{55}Mn , ^{57}Fe , ^{87}Sr , ^{206}Pb , ^{207}Pb , ^{208}Pb and ^{238}U . Data were processed and maps were constructed using HDIP v.1.8 (element mass fractions) and Lolite 4 (U/Pb isotopic ratios, Paton *et al.* 2011). Details on the instrumentation and analytical procedure are described in Appendix S1.

Sr isotopes

Strontium isotopic ratios were determined using both solution-based and *in situ* techniques at the University of Geneva and ETH Zürich, respectively.

MC-ICP-MS: Strontium radiogenic isotope ratios ($^{87}\text{Sr}/^{86}\text{Sr}$) were measured on five aliquots of nanoparticulate powder using a Thermo Neptune PLUS Multi-Collector ICP-MS at the Department of Earth Sciences, University of Geneva, Switzerland, following the methodology of Chiaradia *et al.* (2020). The full analytical procedure is described in Appendix S1.

LA-MC-ICP-MS: *In situ* Sr isotope measurements of experimental charges were undertaken on an ASI (now Applied Spectra) RESolution-SE 193 nm excimer laser ablation system interfaced to a Nu Plasma II MC-ICP-MS at ETH Zürich and following a similar approach to that of Fitzpayne *et al.* (2023) and Tavazzani *et al.* (2024). Full method description is available in Appendix S1.

U-Pb isotopes

Isotopic ratios of U and Pb were determined using both solution-based and *in situ* techniques at ETH Zürich.

ID-TIMS: Five aliquots of 2.1 to 2.4 mg of nanoparticulate powder were dissolved in 6 mol l⁻¹ HCl, equilibrated with the EARTHTIME ^{205}Pb - ^{233}U - ^{235}U tracer solution (Condon *et al.* 2015, Mclean *et al.* 2015) and analysed by isotope dilution thermal ionisation mass spectrometry (ID-TIMS) at ETH Zürich, following the same analytical procedure as in Guillong *et al.* (2024). Detailed description of the methodology can be found in Appendix S1.

LA-ICP-MS: U-Pb measurements in the nanoparticulate pressed powder pellet and experimental charges were performed at ETH Zürich using an ASI (now Applied Spectra) RESolution-LR 193 nm excimer laser ablation system coupled to a Thermo Element XR single-collector sector field ICP-MS

following the workflow of Guillong *et al.* (2024). All raw isotopic ratios were corrected following the standard procedure for carbonate geochronology (Roberts *et al.* 2017, Guillong *et al.* 2020). Measured Pb-Pb and U-Pb ratios were initially corrected for drift and down-hole fractionation using the U-Pb data reduction scheme in Lolite 4 software (Paton *et al.* 2010, 2011) with NIST SRM 614 glass as the calibration reference material. The assumed $^{207}\text{Pb}/^{206}\text{Pb}$ composition of NIST SRM 614 is 0.8708 ± 0.0002 and $^{238}\text{U}/^{206}\text{Pb}$ is 0.70488 ± 0.0070 , as calculated from published data (Jochum *et al.* 2011). Drift-corrected and normalised Pb-isotope ratios obtained in Lolite 4 were then exported offline for further data reduction. Using the U-Pb age of calcite WC-1 (Roberts *et al.* 2017) a correction factor was calculated for the $^{238}\text{U}/^{206}\text{Pb}$ ratios in each measurement session, which corresponds to the quotient of the NIST SRM 614-corrected $^{238}\text{U}/^{206}\text{Pb}$ and the $^{238}\text{U}/^{206}\text{Pb}$ calculated from the published U-Pb age. The nanoparticulate pressed powder pellet $^{238}\text{U}/^{206}\text{Pb}$ ratios were normalised to NIST SRM 614, which provides a closer matrix matching in terms of ablation rates than crystalline carbonate (Wu *et al.* 2018, Jochum *et al.* 2019). Due to the heterogeneity in U/Pb ratios of WC-1 and NIST SRM 614 (Guillong *et al.* 2020) and known long-term excess variance (Sliwinski *et al.* 2022), 2% and 0.5% “external” uncertainty were respectively added in quadrature to the mean $^{238}\text{U}/^{206}\text{Pb}$ and $^{207}\text{Pb}/^{206}\text{Pb}$ ratios obtained from each experimental charge. Detailed description of the instrumental set-up and methodology used is included in Appendix S1.

Results

Physical description

The IHPV experiments investigated the effects of varying sintering temperature (550, 650 and 800 °C) and run time (4 and 24 h) at a constant pressure of 500 MPa (Table 1). Overall, the IHPV experiments produced a crystalline, fine-grained sintered carbonate material. SEM imaging was

performed on a slice of each experimental run (Figure 2), with EBSD analyses to obtain quantitative constraints on grain size (Figure 3) performed on two experiments at 550 °C (SC-D, SC-E) and one experiment each at 650 °C (SC-G) and 800 °C (SC-A).

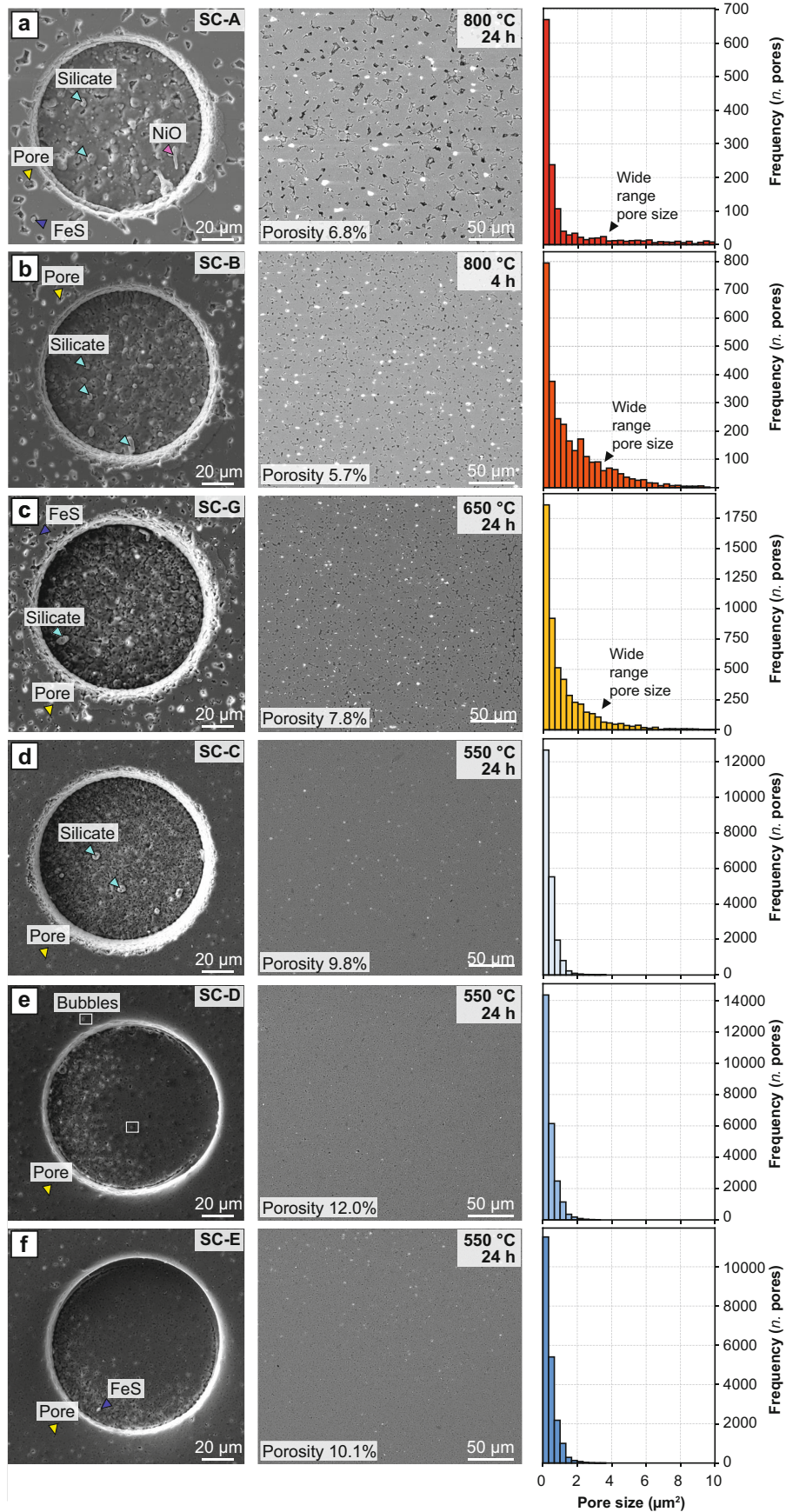
The 550 °C series (SC-C, SC-D, SC-E) is characterised by randomly oriented carbonate grains showing incipient recrystallisation, as indicated by both 120° grain contacts and irregular grain surfaces. Grain cross-sectional areas range up to $30 \mu\text{m}^2$, with 70–75% of grains being smaller than $5 \mu\text{m}^2$ (Figure 3c). The material exhibits a mean pore cross-sectional area of $2 \mu\text{m}^2$ and a total porosity of 10 area % (Figure 2d, e, f). Notably, image analysis of two selected 550 °C experiments yielded reproducible results (compare Figure 3c).

At 650 °C (SC-G), calcite grains exhibit clear recrystallisation (Figure 3b). Grain contacts show equilibrium 120° triple junctions. The grain cross-sectional area ranges up to $90 \mu\text{m}^2$, with approximately 60% of grains in the range of $10\text{--}30 \mu\text{m}^2$ (Figure 3c). The grains remain randomly oriented. The material porosity decreased to 7.8 area%, while the pore cross-sectional area increased to $6 \mu\text{m}^2$ (Figure 2c).

At 800 °C (SC-A, SC-B), regardless of the experimental run time (i.e., 4 h or 24 h), the material is characterised by a homogeneous grain size distribution, with grain cross-sectional areas reaching up to $300 \mu\text{m}^2$ (Figure 3c). The grain orientation remains random, and grain contacts exhibit 120° angles. The porosity decreased to 5.7–6.8 area%, while the pore cross-sectional area increased to $10 \mu\text{m}^2$.

All experiments contain a small fraction of refractory oxides, sulfides and calc-silicates. These non-carbonate mineral phases appear fine-grained (1 to $5 \mu\text{m}^2$ cross-sectional area) in the 550 °C experiments but recrystallise into larger euhedral grains (i.e., $> 10 \mu\text{m}^2$ cross-sectional area), often within the pore space, in higher-temperature experiments.

Figure 2. Texture and porosity characterisation of experimental runs from SEM images. Left: High-magnification secondary electron (SE) images showcase the geometry of ablation crater wall and bottom. These images also include identification of non-carbonate phases, generally $1\text{--}10 \mu\text{m}$ in size, which include sulfides (e.g., FeS), oxides (e.g., NiO) and silicate phases. Centre: Lower-magnification SE images display the varying porosity and pore size characteristics of each experimental charge. These images are used to calculate porosity (area%) and pore size distribution (μm^2). Right: Histograms represent the output of pore size distribution calculations; substantial decrease in maximum pore size and tighter size distributions are observed in experiments conducted at 550 °C (i.e., SC-C, SC-D, SC-E).



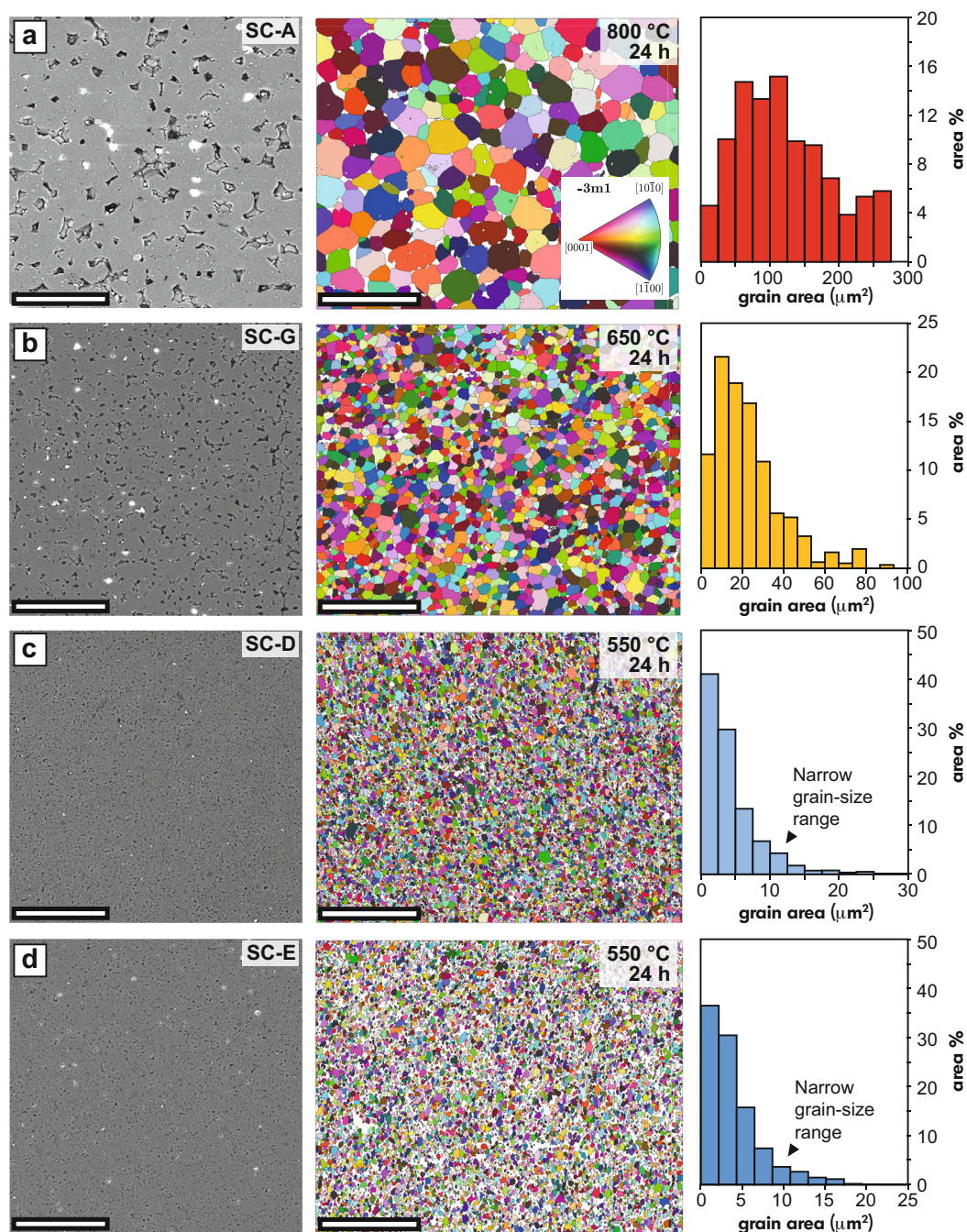


Figure 3. Grain size of experimental charges determined via electron backscatter diffraction (EBSD). Left: Secondary electron images display shape and size of calcite grains. Centre: EBSD grain mean orientation maps coloured for the y-vector direction with inverse pole figure display the orientation and grain boundaries of individual calcite grains. In this plane of view the y-vector is vertical, the x-axis is to the east, and the z-axis is into the plane. The experimental charges have been cut along the smallest axis, meaning that the y-vector corresponds to one of the short axes of the experimental capsule. Note the random orientation of grains and absence of any crystallographic preferred orientation (CPO) across all experimental charges. Right: Histograms display grain size distributions calculated from EBSD grain maps. Maximum grain size area decreases tenfold between experiments run at 800 °C (SC-A) and experiments run at 550 °C (i.e., SC-D, SC-E). Scale bar across panels is 50 μm .

Major and trace element distribution

EPMA analysis of one 550 °C experimental charge (SC-E) yielded a CaO fraction of $53.36 \pm 1.9\%$ *m/m* ($n = 21$, 2s), equivalent to a Ca mass fraction of $381324 \pm 7387 \mu\text{g g}^{-1}$. EPMA analysis of the 4 h run at 800 °C (SC-B) yielded a comparable CaO value of $53.41 \pm 2.4\%$ *m/m* ($n = 30$, 2s), equivalent to a Ca mass fraction of $381700 \pm 9157 \mu\text{g g}^{-1}$. Individual measurements were performed across the surface area of a slice across the experimental charge of interest, with no systematic compositional trends detected. Given the limited overall uncertainty (ca. 2% relative standard error) of the Ca mass fraction obtained by EPMA analyses across both experimental results, we use the mean mass fraction of both experimental slices of $381545 \pm 8402 \mu\text{g g}^{-1}$ ($n = 51$, 2s) as the internal reference during data reduction for subsequent LA-ICP-MS trace element measurements. Other minor elements quantified during EPMA analyses (Si, Mg, Fe, Na) show heterogeneity at the sampling scale ($< 10 \mu\text{m}$) across both experimental charges, with the 800 °C experiment (SC-B) characterised by a more marked compositional scatter (e.g., Fe and Na, see Appendix S2).

A set of forty-three minor and trace elements was quantified by LA-ICP-MS (Table 2) on the pressed powder pellet and on one slice for each experimental charge, to test for the degree of compositional homogeneity across individual experiments and existing compositional variations between starting nanoparticulate powder and sintering experiments. The results show that most quantified elements are characterised by limited relative standard deviation (5–10% RSD) within individual experiments (Figure 4), with the notable exception of a suite of chalcophile elements (e.g., Co, Ni, Cu, Zn, Mo) and high field-strength elements (i.e., Ti, Zr, Nb, Hf) showing large to very large variations in individual experiments (10–30% RSD). Noteworthy is the variation in mass fraction for selected elements between starting nanoparticulate powder and experimental charges (Figure 5). The 800 °C experiments show depletion, up to 30% from the original content, in a suite of volatile (i.e., Li, Na, Mg, Al, K) and chalcophile elements (e.g., Fe, Co, Cu, Zn, Mo, Pb). This is paralleled by enrichments in Ni, Zr and Hf. Depletion, albeit to a lesser degree, in a similar set of elements is also noticeable in experiments run at 650 and 550 °C.

The LA-ICP-MS analyses indicate a reproducible Sr mass fraction across experimental charges (mean of all experiments $7442 \pm 259 \mu\text{g g}^{-1}$, $n = 320$, 2s), slightly lower than the mean of analyses on the pressed powder pellet (7796

$\pm 218 \mu\text{g g}^{-1}$, $n = 30$, 2s; Figure 6a). Rubidium, the main interfering element for *in situ* Sr isotope measurements, yielded a mean value across all experimental charges of $3.3 \pm 1.0 \mu\text{g g}^{-1}$ ($n = 320$, 2s; Figure 6b), resulting in a mean Rb/Sr ratio of 0.0004. Other potential interfering elements are also at very low mass fractions (Dy = $0.29 \pm 0.04 \mu\text{g g}^{-1}$; Er = $0.21 \pm 0.03 \mu\text{g g}^{-1}$; Yb = $0.20 \pm 0.03 \mu\text{g g}^{-1}$, 2s, $n = 320$; Figure 6c) and should not be considered problematic for accurate *in situ* Sr isotopic determinations (Weber *et al.* 2018).

Mass fractions of U and Pb are also consistent across experimental charges with a slightly lower Pb content observed in the 24 h IHPV run at 800 °C (Figure 6d). The mean U mass fraction across all experimental charges is $2.3 \pm 0.4 \mu\text{g g}^{-1}$ (2s, $n = 320$), while Pb yielded a mean mass fraction value of $1.4 \pm 0.5 \mu\text{g g}^{-1}$ (2s, $n = 320$; quantified based on measured ^{208}Pb mass). These values are overlapping within uncertainty to slightly lower than measured mass fractions of the pressed powder pellet (Figure 5).

Trace element imaging

Imaged element mass fractions for three experimental charges (i.e., SC-A, SC-B and SC-C) are in general agreement with elemental mass fractions obtained by spot measurements (Figure 7, Figure S3). The mass fraction of Sr does not show any significant spatial variability in SC-A and SC-B. In SC-C, Sr mass fraction variability is observed in the surroundings of fractures, which are characterised by slightly higher contents (up to $9000 \mu\text{g g}^{-1}$). Other measured minor elements (e.g., Mg, Mn, Fe) display near-homogenous distribution in SC-A and SC-B with slight local enrichments observed in SC-C in proximity of fractures. The outer rim of all experimental charges (i.e., volume of the experiments in direct contact with silver capsule) is enriched in Pb (up to $8 \mu\text{g g}^{-1}$, Figure 7) compared with the internal experimental volumes, which instead show remarkable homogeneity. The thickness of the Pb-enriched rim is 100–200 μm in SC-A, 100 μm in SC-B and $< 50 \mu\text{m}$ in SC-C. A few small areas (tens of μm) showing higher Pb mass fraction can also be identified in SC-C (Figure 7c). The mass fraction of U, while homogeneously distributed in the 24 h experimental runs (SC-A and SC-C), shows substantial spatial variations in the 4 h run (SC-B) with areas of higher U mass fraction ($4\text{--}5 \mu\text{g g}^{-1}$) surrounded by a network displaying mean U mass fraction ($2.5\text{--}3.0 \mu\text{g g}^{-1}$, Figure 7b). Moreover, higher temperature experimental charges (SC-A, SC-B) are characterised by a slight depletion in U in the outer rim (i.e., in a band 50–100 μm wide, Figure 7a, b).

Table 2.

Compilation of trace element data ($\mu\text{g g}^{-1}$), including mean (preferred mean*) and uncertainty (1s) for the analysed experimental material

	Nano-powder	SC-A (800 °C)	SC-B (800 °C)	SC-C (550 °C)	SC-D (550 °C)	SC-E (550 °C)	SC-G (650 °C)
⁷ Li	4.0 ± 0.4	2.5 ± 0.1	3.0 ± 0.2	3.4 ± 0.2	4.0 ± 0.2	3.8 ± 0.1	3.3 ± 0.1
²³ Na	606 ± 45	509 ± 36	562 ± 37	629 ± 36	677 ± 42	654 ± 37	622 ± 27
²⁴ Mg	2186 ± 44	1961 ± 46	1982 ± 29	2030 ± 28	2068 ± 42	2071 ± 27	2057 ± 38
²⁷ Al	2376 ± 74	1893 ± 152	2104 ± 125	2165 ± 49	2189 ± 64	2222 ± 73	2188 ± 65
²⁹ Si	9045 ± 306	8143 ± 249	8558 ± 251	8333 ± 270	8524 ± 263	8475 ± 306	8943 ± 322
³⁹ K	667 ± 28	591 ± 44	560 ± 75	569 ± 31	597 ± 27	583 ± 34	585 ± 20
⁴⁵ Sc	0.58 ± 0.04	0.57 ± 0.04	0.63 ± 0.03	0.65 ± 0.02	0.66 ± 0.03	0.68 ± 0.03	0.67 ± 0.03
⁴⁹ Ti	104 ± 9	97 ± 15	100 ± 9	98 ± 8	104 ± 12	100 ± 10	103 ± 13
⁵¹ V	4.8 ± 0.1	4.5 ± 0.1	4.5 ± 0.1	4.6 ± 0.1	4.7 ± 0.1	4.7 ± 0.1	4.6 ± 0.1
⁵³ Cr	2.5 ± 0.1	2.3 ± 0.3	2.3 ± 0.2	2.3 ± 0.3	2.4 ± 0.1	2.4 ± 0.2	2.3 ± 0.2
⁵⁵ Mn	262 ± 4	246 ± 3	250 ± 2	251 ± 2	254 ± 3	254 ± 2	254 ± 2
⁵⁷ Fe	4132 ± 109	3461 ± 200	3674 ± 108	3873 ± 92	3880 ± 93	3839 ± 80	3799 ± 204
⁵⁹ Co	0.80 ± 0.06	0.62 ± 0.10	0.79 ± 0.09	0.79 ± 0.10	0.77 ± 0.12	0.85 ± 0.16	0.73 ± 0.16
⁶⁰ Ni	1.9 ± 0.5	1.6 ± 0.4	2.5 ± 0.3	6.4 ± 0.6	1.9 ± 0.3	2.0 ± 0.2	1.9 ± 0.2
⁶⁵ Cu	2.6 ± 0.6	1.7 ± 0.7	1.6 ± 0.2	1.8 ± 0.2	2.2 ± 0.6	2.7 ± 1.2	1.6 ± 0.2
⁶⁶ Zn	5.0 ± 1.6	4.3 ± 0.4	2.7 ± 0.2	2.8 ± 0.4	2.7 ± 0.3	2.4 ± 0.2	3.2 ± 0.4
⁸⁵ Rb	3.5 ± 0.1	3.4 ± 0.3	2.7 ± 0.7	3.0 ± 0.1	3.1 ± 0.1	3.1 ± 0.1	2.8 ± 0.1
⁸⁸ Sr	7796 ± 109	7392 ± 57	7369 ± 43	7425 ± 54	7456 ± 77	7493 ± 55	7395 ± 60
⁸⁹ Y	2.73 ± 0.07	2.73 ± 0.05	2.74 ± 0.03	2.68 ± 0.05	2.66 ± 0.04	2.71 ± 0.04	2.71 ± 0.06
⁹⁰ Zr	3.1 ± 0.6	4.3 ± 0.3	3.8 ± 0.3	3.0 ± 0.7	3.1 ± 1.0	3.2 ± 0.9	3.0 ± 0.5
⁹³ Nb	0.31 ± 0.03	0.27 ± 0.05	0.30 ± 0.04	0.29 ± 0.02	0.31 ± 0.05	0.29 ± 0.03	0.30 ± 0.04
⁹⁵ Mo	0.15 ± 0.03	0.11 ± 0.02	0.10 ± 0.02	0.10 ± 0.01	0.14 ± 0.03	0.10 ± 0.01	0.12 ± 0.03
¹³³ Cs	0.22 ± 0.01	0.22 ± 0.03	0.17 ± 0.07	0.15 ± 0.01	0.17 ± 0.02	0.16 ± 0.01	0.14 ± 0.01
¹³⁷ Ba	146 ± 5	142 ± 2	141 ± 3	138 ± 5	139 ± 5	140 ± 7	135 ± 3
¹³⁹ La	1.22 ± 0.10	1.31 ± 0.05	1.26 ± 0.06	1.17 ± 0.09	1.18 ± 0.10	1.19 ± 0.10	1.23 ± 0.06
¹⁴⁰ Ce	2.24 ± 0.18	2.35 ± 0.10	2.27 ± 0.10	2.13 ± 0.20	2.17 ± 0.20	2.18 ± 0.20	2.20 ± 0.10
¹⁴¹ Pr	0.28 ± 0.02	0.29 ± 0.01	0.27 ± 0.01	0.27 ± 0.02	0.27 ± 0.02	0.28 ± 0.02	0.28 ± 0.01
¹⁴⁶ Nd	1.20 ± 0.08	1.26 ± 0.05	1.23 ± 0.05	1.15 ± 0.06	1.16 ± 0.09	1.17 ± 0.07	1.18 ± 0.05
¹⁴⁷ Sm	0.26 ± 0.01	0.27 ± 0.02	0.27 ± 0.01	0.25 ± 0.02	0.25 ± 0.02	0.26 ± 0.02	0.26 ± 0.01
¹⁵¹ Eu	0.063 ± 0.003	0.064 ± 0.002	0.062 ± 0.003	0.060 ± 0.003	0.060 ± 0.003	0.061 ± 0.003	0.062 ± 0.003
¹⁵⁷ Gd	0.303 ± 0.016	0.301 ± 0.010	0.297 ± 0.013	0.292 ± 0.009	0.287 ± 0.016	0.289 ± 0.014	0.296 ± 0.012
¹⁵⁹ Tb	0.044 ± 0.003	0.044 ± 0.002	0.043 ± 0.002	0.044 ± 0.002	0.043 ± 0.002	0.043 ± 0.002	0.044 ± 0.002
¹⁶³ Dy	0.304 ± 0.017	0.302 ± 0.010	0.295 ± 0.008	0.294 ± 0.009	0.287 ± 0.011	0.288 ± 0.012	0.293 ± 0.011
¹⁶⁵ Ho	0.067 ± 0.003	0.066 ± 0.002	0.066 ± 0.003	0.066 ± 0.002	0.064 ± 0.002	0.065 ± 0.002	0.066 ± 0.003
¹⁶⁷ Er	0.212 ± 0.014	0.212 ± 0.008	0.213 ± 0.008	0.200 ± 0.007	0.205 ± 0.010	0.207 ± 0.009	0.210 ± 0.009
¹⁶⁹ Tm	0.03 ± 0.002	0.029 ± 0.001	0.029 ± 0.001	0.029 ± 0.002	0.029 ± 0.001	0.029 ± 0.002	0.030 ± 0.001
¹⁷³ Yb	0.200 ± 0.013	0.208 ± 0.009	0.213 ± 0.009	0.206 ± 0.009	0.198 ± 0.009	0.201 ± 0.009	0.203 ± 0.007
¹⁷⁵ Lu	0.031 ± 0.002	0.032 ± 0.001	0.032 ± 0.001	0.031 ± 0.001	0.031 ± 0.002	0.031 ± 0.001	0.031 ± 0.002
¹⁷⁸ Hf	0.072 ± 0.017	0.104 ± 0.009	0.092 ± 0.010	0.073 ± 0.017	0.070 ± 0.019	0.073 ± 0.019	0.072 ± 0.013
¹⁸¹ Ta	0.020 ± 0.002	0.019 ± 0.004	0.020 ± 0.003	0.021 ± 0.003	0.020 ± 0.003	0.020 ± 0.002	0.020 ± 0.003
²⁰⁸ Pb	1.67 ± 0.09	1.23 ± 0.19	1.50 ± 0.05	1.47 ± 0.05	1.48 ± 0.05	1.47 ± 0.04	1.48 ± 0.03
²³² Th	0.279 ± 0.016	0.285 ± 0.019	0.292 ± 0.013	0.289 ± 0.014	0.283 ± 0.021	0.283 ± 0.015	0.295 ± 0.009
²³⁸ U	2.37 ± 0.04	2.24 ± 0.11	2.29 ± 0.18	2.30 ± 0.04	2.25 ± 0.05	2.27 ± 0.05	2.14 ± 0.11

* Preferred mean = mean of analysis within ± 1 standard deviation of the overall mean.

Sr isotopic measurements

The ⁸⁷Sr/⁸⁶Sr ratios obtained by solution MC-ICP-MS yielded a mean ratio of 0.708104 ± 0.000002 (2s, $n = 28$, Table 3) on nanoparticulate powder obtained from the natural carbonate. The mean ⁸⁴Sr/⁸⁶Sr ratio of all measurements is 0.056484 ± 0.000002 (2s, $n = 28$, Table 3). The full solution data set including individual measurements is included in Appendix S4.

LA-MC-ICP-MS Sr isotope measurements were performed both on the pressed powder pellet and for all experimental charges to verify homogeneity between the starting material and the sintered calcite. In each experimental slice, forty analyses (38 μm spot size) randomly distributed across the surface ensured representative sampling. The measured ⁸⁷Sr/⁸⁶Sr ratios show larger variability in comparison with the solution data (Figure 8); however, the mean ⁸⁷Sr/⁸⁶Sr ratio of individual experimental products

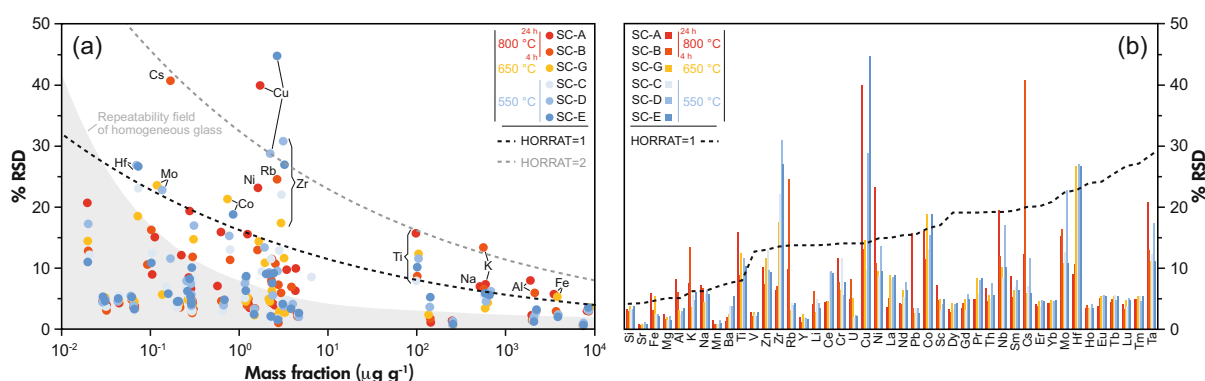


Figure 4. (a) Relative standard deviation (% RSD) against mean mass fraction in $\mu\text{g g}^{-1}$ (log scale) for forty spot measurements ($43 \mu\text{m}$ spot size) on each experimental charge. Several elements display HORRAT (Horwitz ratio, Horwitz and Albert 1995) above 1, thus indicating insufficient reproducibility at the experiment level. These elements are labelled individually or in groups. In grey the repeatability field of LA-ICP-MS analyses of homogenous glasses (Jochum *et al.* 2011, 2019). (b) Inter-experimental variation of all determined elements displayed as RSD of forty spot measurements ($43 \mu\text{m}$ spot size), compared with expected (maximum) RSD calculated after Horwitz and Albert (1995). Results for individual experimental charges are shown in Figure S4.

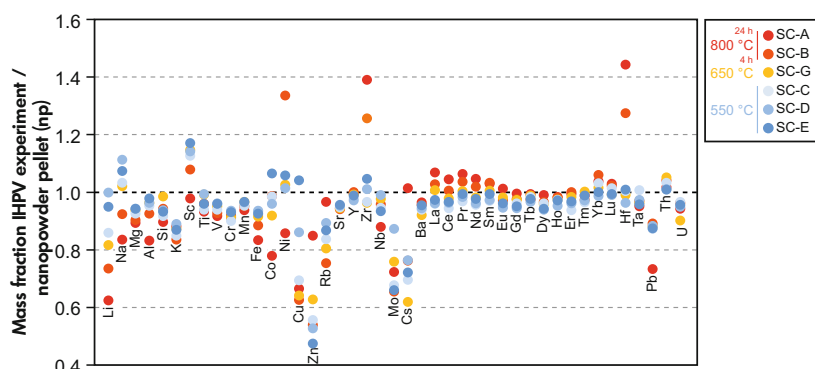


Figure 5. Ratio of mass fraction of experimental charges and original nanoparticulate powder (i.e., analysed as pressed pellet). Enrichment in Hf and Zr is prominent in higher temperature experiments, while Na and Sc are slightly enriched in lower temperature experimental charges. Moderate to significant depletions are observed for selected lithophile (e.g., Li, Na, Al, Rb, Cs), siderophile (e.g., Mo, Co) and chalcophile (e.g., Cu, Zn, Pb) elements. Depletion in these elements is generally more pronounced for experiments carried out at higher temperatures. Outside of these notable exceptions, most determined elements, including key elements such as U, REE, Y and Sr, are within 5–10% of the starting nanoparticulate powder mass fraction. Mean of forty measurements is plotted. Uncertainty on individual mean values is equivalent to relative standard deviation in Figure 4.

and the pressed powder pellet agree within uncertainty with the solution data. The mean $^{84}\text{Sr}/^{86}\text{Sr}$ ratios measured for individual experiments are generally within uncertainty of the solution value, with three experiments (SC-A, SC-D and SC-E) yielding slightly offset mean values (Figure 8). Overall, no correlation between experimental conditions (i.e., temperature or experimental run duration) and $^{87}\text{Sr}/^{86}\text{Sr}$ or $^{84}\text{Sr}/^{86}\text{Sr}$ ratios are observed, similarly to the lack of substantial variation in Sr mass fraction or mass fraction of

interfering elements between experimental charges (Table 2, Figure 6).

U-Pb isotopic composition

The U-Pb isotopic measurements by ID-TIMS on five aliquots of starting carbonate nanoparticulate powder yielded a mean $^{238}\text{U}/^{206}\text{Pb}$ ratio of 4.6244 ± 0.0057

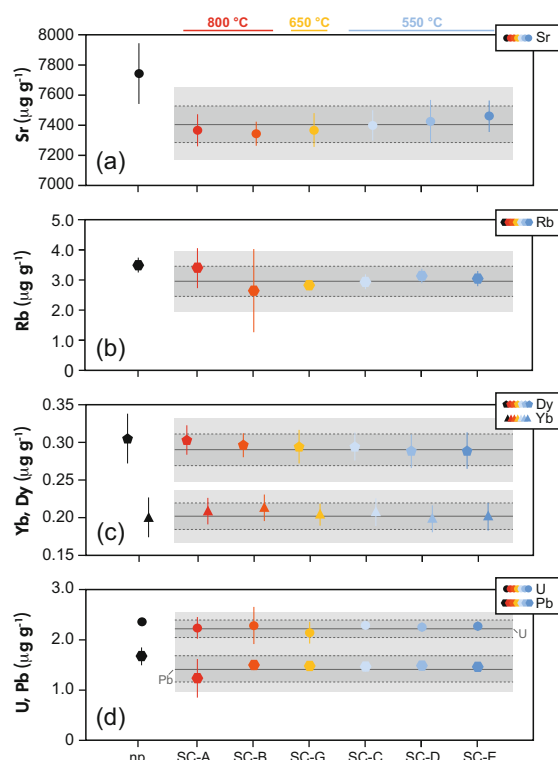


Figure 6. Mass fraction of elements key for *in situ* measurements of Sr and U/Pb isotope ratios. Mean mass fraction of (a) Sr, (b) Rb, (c) potential interfering REE for Sr isotopic measurements, (d) U and Pb for each experimental charge compared with the mean for starting nanoparticulate powder (black symbol). Uncertainties are 2s for individual samples. Solid horizontal lines represent the mean of all analyses on experimental charges including 1 s (dashed lines) and 2s envelopes.

(2s, $n = 5$) and a $^{207}\text{Pb}/^{206}\text{Pb}$ ratio of 0.66226 ± 0.00016 (2s, $n = 5$) (Table 4, Figure 9g). The full ID-TIMS dataset including individual measurements and all other U/Pb ratios is included in Appendix S4.

Three slices cut across the length of each experimental capsule (Figure 1d) were analysed by LA-ICP-MS to test for inter- and intra-experimental U/Pb isotopic homogeneity. A nanoparticulate pressed powder pellet was also analysed in order to check U-Pb isotopic equivalence between starting nanoparticulate powder and sintered calcite. Overall, U/Pb isotopic ratios show increasing divergence from isotopic composition of the starting material and increasing internal heterogeneity with higher sintering temperature (Table 4, Figure 7, Figure 9).

Analyses of the pressed powder pellet yielded a mean $^{207}\text{Pb}/^{206}\text{Pb}$ ratio of 0.6653 ± 0.0035 (2s, $n = 41$) and mean $^{238}\text{U}/^{206}\text{Pb}$ ratio of 4.667 ± 0.095 (2s, $n = 41$). These ratios are within uncertainty of the mean ratios obtained with solution analysis of starting carbonate nanoparticulate powder (Figure 9g).

The 550 °C series experiments yielded nearly identical mean $^{207}\text{Pb}/^{206}\text{Pb}$ ratios (SC-C: 0.6643 ± 0.0033 , 2s, $n = 348$; SC-D: 0.6653 ± 0.0034 , 2s, $n = 100$; SC-E: 0.6636 ± 0.0034 , 2s, $n = 150$) and mean $^{238}\text{U}/^{206}\text{Pb}$ ratios (SC-C: 4.554 ± 0.092 , 2s, $n = 348$; SC-D: 4.524 ± 0.094 , 2s, $n = 100$; SC-E: 4.596 ± 0.093 , 2s, $n = 150$). Internal homogeneity of each experimental run is attested by the low RSD of both $^{207}\text{Pb}/^{206}\text{Pb}$ and $^{238}\text{U}/^{206}\text{Pb}$ ratios ($= 0.5$ to 0.6% RSD for $^{207}\text{Pb}/^{206}\text{Pb}$ and $= 2.1$ to 2.8% RSD for $^{238}\text{U}/^{206}\text{Pb}$). These ratios are within uncertainty of the mean ratios obtained with solution analysis of starting carbonate nanoparticulate powder (Figure 9g). Imaging of one 550 °C experimental charge (SC-C, Figure 7c) confirms the homogenous distribution of U/Pb ratios across this series of experiments, only a thin (tens of μm) rim towards the edge of the capsule displays slightly lower $^{238}\text{U}/^{206}\text{Pb}$ values.

The experiment conducted at 650 °C (SC-G) yielded mean $^{207}\text{Pb}/^{206}\text{Pb}$ (0.6647 ± 0.0033 , 2s, $n = 178$) and $^{238}\text{U}/^{206}\text{Pb}$ (4.447 ± 0.092 , 2s, $n = 178$) ratios comparable to those obtained for the lower temperature experiments. However, the increased isotopic heterogeneity of the 650 °C experimental charge is apparent in the higher RSD observed, specifically for the $^{238}\text{U}/^{206}\text{Pb}$ ratio (3.5%).

At 800 °C, the shorter experimental run (4 h, SC-B) displays a substantial offset in mean $^{207}\text{Pb}/^{206}\text{Pb}$ ratio (0.6748 ± 0.0052 , 2s, $n = 162$) and a significant difference in mean $^{238}\text{U}/^{206}\text{Pb}$ ratio (4.120 ± 0.184 , 2s, $n = 162$) compared with the mean isotopic ratios of both the pressed powder pellet values and lower temperature experiments. The poor isotopic homogeneity of the material is reflected in the large RSD, one order of magnitude higher than in the lower temperature experiments (3.7% for $^{207}\text{Pb}/^{206}\text{Pb}$ and 14.3% for $^{238}\text{U}/^{206}\text{Pb}$). The longer experimental run (24 h, SC-A) displays an even larger isotopic offset from the pressed powder pellet and lower temperature experiments than the shorter runs, especially evident in the Pb ratios ($^{207}\text{Pb}/^{206}\text{Pb}$ of 0.6953 ± 0.0061 , 2s, $n = 148/150$; $^{238}\text{U}/^{206}\text{Pb}$ of 4.683 ± 0.234 , 2s, $n = 148/150$). Moreover, this experiment shows the highest RSD values (4.5% for $^{207}\text{Pb}/^{206}\text{Pb}$ and 33.0% for $^{238}\text{U}/^{206}\text{Pb}$). Mapping and spot analyses across the 800 °C experimental capsules

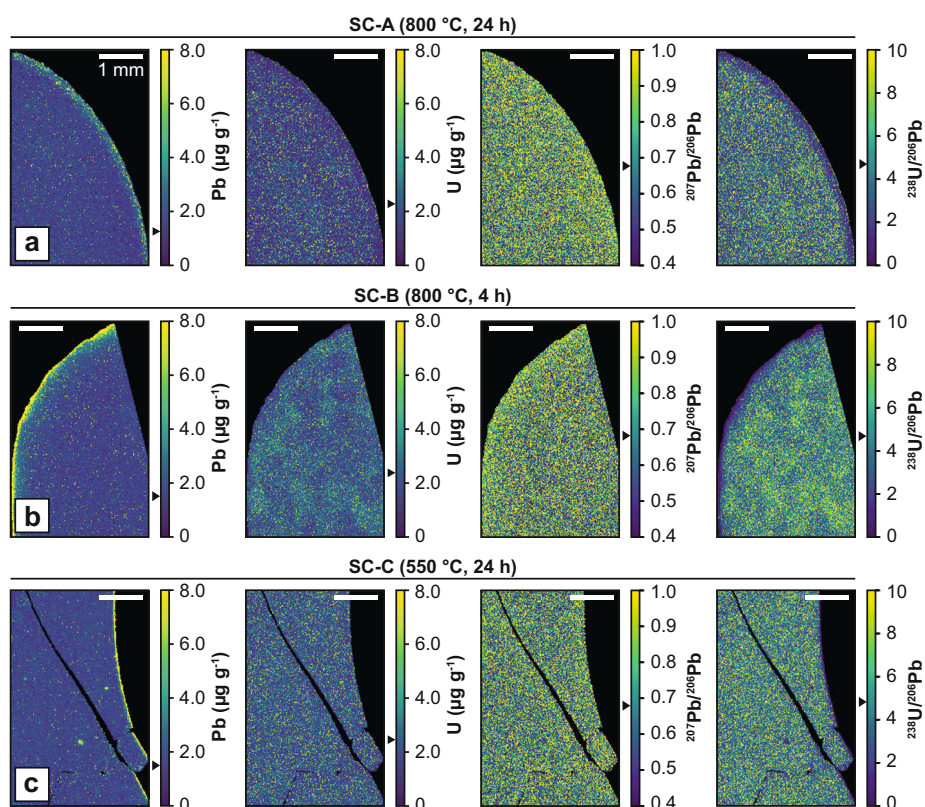


Figure 7. U/Pb isotopes and element distribution maps (Pb, U) for three experimental charges. Approximately one quarter of the surface area of (a) SC-A, (b) SC-B and (c) SC-C was analysed (see Figure S2 for location of analyses in experimental charge). The black triangle next to colour scale indicates the mean isotopic ratio and element mass fraction obtained from LA-ICP-MS static spot analyses (Table 2).

Table 3.
 $^{87}\text{Sr}/^{86}\text{Sr}$ and $^{84}\text{Sr}/^{86}\text{Sr}$ ratios obtained for different analytical techniques and analysed materials

Material	Analytical technique	$^{87}\text{Sr}/^{86}\text{Sr} (\pm 2s^a)$	$^{84}\text{Sr}/^{86}\text{Sr} (\pm 2s)$	No. of analyses
Nano-powder	MC-ICP-MS	0.7081041 ± 0.0000016	0.056484 ± 0.000002	28
Nano-powder	LA-MC-ICP-MS	0.708133 ± 0.000053	0.05636 ± 0.00015	40
SC-A (800 °C)	LA-MC-ICP-MS	0.708085 ± 0.000054	0.05626 ± 0.00018	39
SC-B (800 °C)	LA-MC-ICP-MS	0.708109 ± 0.000053	0.05634 ± 0.00015	40
SC-G (650 °C)	LA-MC-ICP-MS	0.708115 ± 0.000053	0.05639 ± 0.00017	40
SC-C (550 °C)	LA-MC-ICP-MS	0.708128 ± 0.000053	0.05642 ± 0.00014	40
SC-D (550 °C)	LA-MC-ICP-MS	0.708155 ± 0.000053	0.05625 ± 0.00015	40
SC-E (550 °C)	LA-MC-ICP-MS	0.708126 ± 0.000053	0.05632 ± 0.00011	40

^a For LA-MC-ICP-MS results, includes "external" uncertainty from validation reference materials.

show a strong spatial control on the distribution of $^{238}\text{U}/^{206}\text{Pb}$ and $^{207}\text{Pb}/^{206}\text{Pb}$ ratios. Both SC-A and SC-B are characterised by steadily increasing $^{207}\text{Pb}/^{206}\text{Pb}$ and anomalously low $^{238}\text{U}/^{206}\text{Pb}$ in a ca. 0.5 mm wide area in proximity of the capsule walls (Figure S6). Imaging of experiment SC-B also highlights the spatial heterogeneity of U/Pb ratios, characterised by patchy distribution of high U/Pb areas (Figure 7b).

Discussion and validation of SC-550

The motivation for this study is to produce a reference material that displays a degree of homogeneity in its U-Pb elemental mass fraction, isotopic ratios and ablation behaviour such that it can be used as matrix-matched RM for U-Pb geochronology of natural carbonate samples. Overall, imaging and *in situ* analysis results show that the

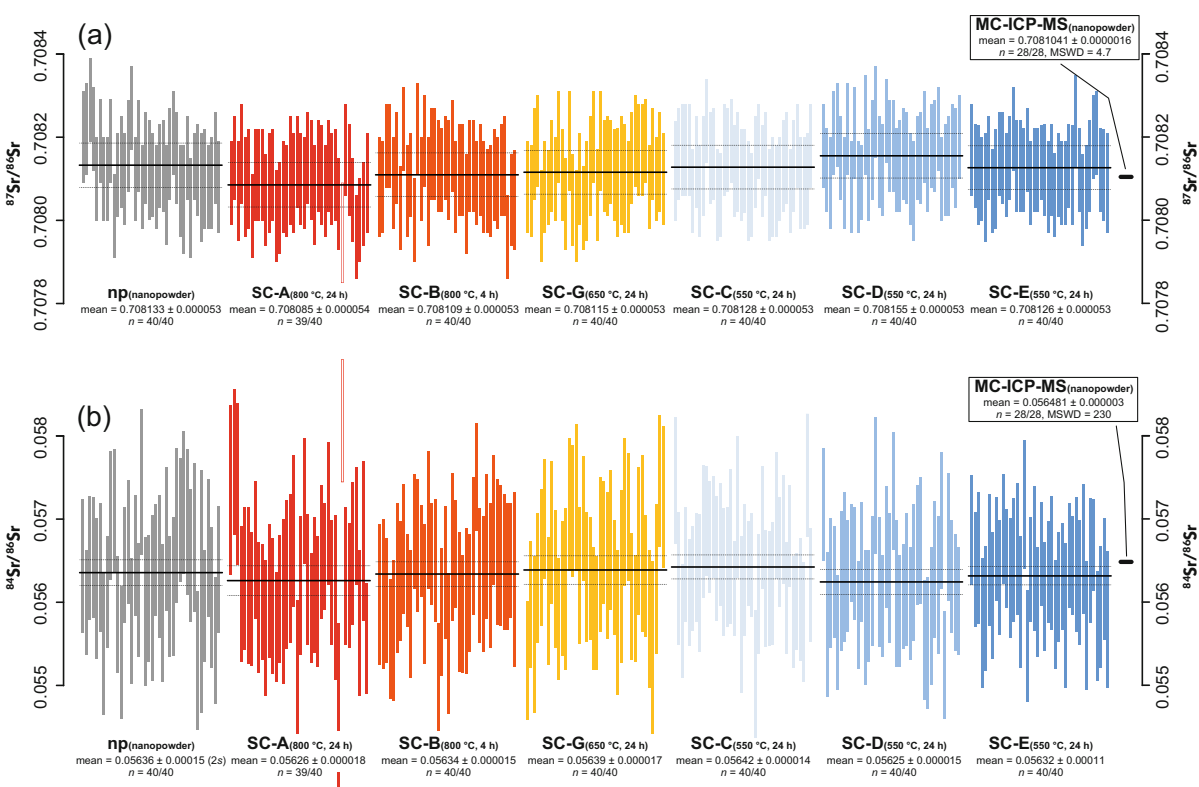


Figure 8. Laser ablation MC-ICP-MS Sr isotope data. For each experimental charge the (a) $^{87}\text{Sr}/^{86}\text{Sr}$ and the (b) $^{84}\text{Sr}/^{86}\text{Sr}$ ratios of forty individual measurements are shown as well as the mean ratio (solid horizontal lines) including 2s uncertainties (dashed lines). Vertical bars represent 2s uncertainty on individual measurements. Thick black lines on the far right of the plot represent the mean $^{87}\text{Sr}/^{86}\text{Sr}$ and $^{84}\text{Sr}/^{86}\text{Sr}$ ratios and corresponding 2s uncertainty from solution MC-ICP-MS measurements ($n = 28$) obtained at the University of Geneva (Table 3).

Table 4. $^{238}\text{U}/^{206}\text{Pb}$ and $^{207}\text{U}/^{206}\text{Pb}$ ratios obtained for the different analytical techniques and analysed materials

Material	Analytical technique	$^{238}\text{U}/^{206}\text{Pb}$ ($\pm 2s^a$)	% RSD	$^{207}\text{Pb}/^{206}\text{Pb}$ ($\pm 2s^b$)	% RSD	No. of analyses
Nano-powder	ID-TIMS	4.6244 ± 0.0057	0.15	0.66226 ± 0.00016	0.04	5
Nano-powder	LA-ICP-MS	4.667 ± 0.095	1.3	0.6653 ± 0.0035	0.6	41
SC-A (800 °C)	LA-ICP-MS	4.683 ± 0.234	33.0	0.6953 ± 0.0061	4.5	148
SC-B (800 °C)	LA-ICP-MS	4.120 ± 0.184	14.3	0.6748 ± 0.0052	3.7	162
SC-G (650 °C)	LA-ICP-MS	4.447 ± 0.092	3.5	0.6647 ± 0.0033	0.5	178
SC-C (550 °C)	LA-ICP-MS	4.554 ± 0.092	2.1	0.6643 ± 0.0033	0.6	348
SC-D (550 °C)	LA-ICP-MS	4.524 ± 0.094	2.8	0.6653 ± 0.0034	0.5	100
SC-E (550 °C)	LA-ICP-MS	4.596 ± 0.093	2.1	0.6636 ± 0.0034	0.5	150

^a For LA-ICP-MS results, includes 2% “external” uncertainty due to U/Pb heterogeneity of the calibration reference material (Guillong *et al.* 2020) and long-term excess variance (Sliwinski *et al.* 2022).

^b For LA-ICP-MS results, includes 0.5% long-term excess variance (Sliwinski *et al.* 2022).

materials resulting from experiments conducted at high temperature (i.e., > 600 °C; SC-A, SC-B, SC-G), regardless of the experimental runtime, are characterised by substantial textural and compositional heterogeneity. Conversely, the experiments ran at 550 °C for 24 h (i.e., SC-C, SC-D, SC-E) produced material with good textural and compositional

homogeneity at the scale of LA-ICP-MS analyses (tens to hundreds of μm). We call this material SC-550 (Sintered Carbonate – 550 °C). Below, we report provisional information values for the use of SC-550 as quality control RM in trace element and Sr isotopic analyses of natural carbonates and provide reference values for U/Pb isotopic

ratios and a comprehensive discussion of its suitability for U-Pb geochronology.

Elemental mass fraction and $^{87}\text{Sr}/^{86}\text{Sr}$ – homogeneity and information values

Point analyses and imaging of SC-550 show a high degree of homogeneity in trace element mass fraction and absolute concentrations that are comparable to that of natural carbonates in a set of key elements (e.g., Ba, Li, Mg, Mn, Sr). Additionally, the high Sr mass fraction (ca. $7500\ \mu\text{g g}^{-1}$) and the low abundance of interfering elements (i.e., REE, Rb) make it a potential candidate for matrix-matched calibration of *in situ* measurements of Sr isotopes in carbonate samples with high-Sr abundance (e.g., corals, molluscs, otoliths).

The level of homogeneity for forty-three elements is tested *in situ* in SC-550 experimental charges for sample portion masses $< 1\ \mu\text{g}$ (Jochum and Enzweiler 2014). Specifically, the RSD values are calculated as the variation between forty static spot ablations across individual experimental charges using a small spot size ($43\ \mu\text{m}$), making the mobilised mass of sintered carbonate per ablation ca. $0.2\ \mu\text{g}$. The RSD values strongly depend on element mass fraction and vary between 2–5% (at mass fractions $> 10\ \mu\text{g g}^{-1}$) and 5–30% (at mass fractions $< 1\ \mu\text{g g}^{-1}$). Generally, RSD values are similar to reported repeatability values for LA-ICP-MS analyses obtained on other microanalytical RMs (e.g., NIST, USGS, MPI-DING glasses, see Figure 4a; Jochum *et al.* 2011, 2019, Boer *et al.* 2021). The elemental mapping of a large surface area ($2.5 \times 5.0\ \text{mm}$) conducted for one of the SC-550 experiments (SC-C) confirms the good micro-homogeneity of the material and reveals only a limited spatial heterogeneity in trace elements, which is limited to the area immediately surrounding (tens of μm) both open and sealed fractures.

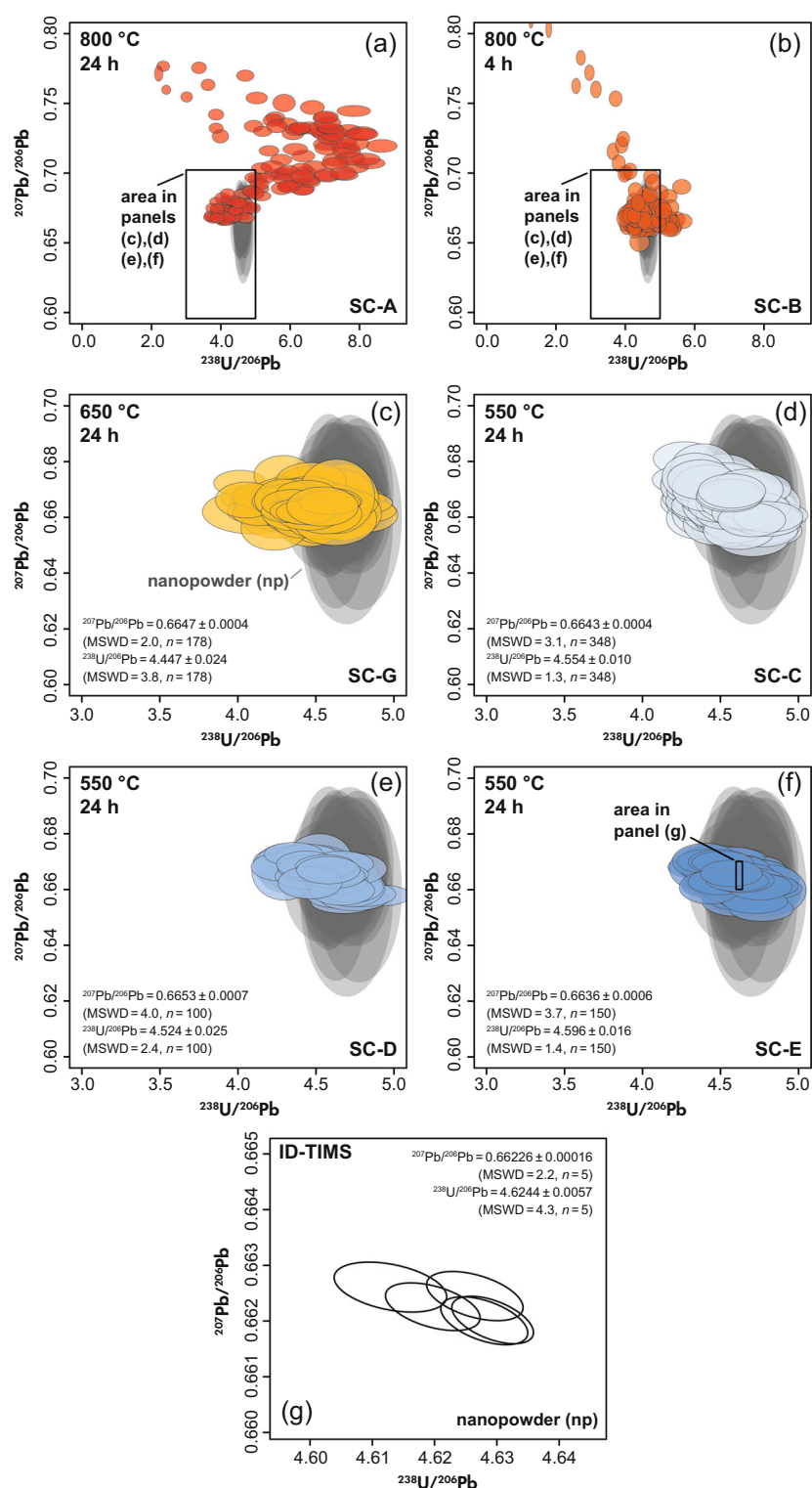
Excluding elements showing % RSD above maximum values expected for a homogenous material (i.e., HORRAT values > 1 , Horwitz and Albert 1995; Figure 4), the preferred mean mass fraction values for individual elements in SC-550 experiments (SC-C, SC-D, SC-E) reported in Table 2 can be used as provisional information value for quality control of LA-ICP-MS trace element measurements in carbonate materials. The suite of elements with poor reproducibility (Ti, Co, Ni, Cu, Rb, Mo, Zr, Hf, Figure 4a) are generally less relevant for applications such as environmental or palaeoclimate research. Notwithstanding the micro-homogeneity of the

material, the element mass fractions of starting nanoparticulate powder and sintered experimental charges need to be calibrated by external methods (i.e., solution analyses) before SC-550 can be used as a matrix matched RM for quantification of element mass fraction in natural carbonates.

The sample used to produce the nanoparticulate powder used in the experimental runs (RA138, Guillong *et al.* 2024) was collected from uppermost Serpukhovian (middle Carboniferous) limestone strata and dated by U-Pb (ID-TIMS) to $321.99 \pm 0.65\ \text{Ma}$. The mean $^{87}\text{Sr}/^{86}\text{Sr}$ ratio of all solution-based measurements on the starting pressed powder pellet is 0.7081041 ± 0.0000016 ($2s$, $n = 28$). This value is in good agreement with the $^{87}\text{Sr}/^{86}\text{Sr}$ ratio of 0.70800–0.70820 for Serpukhovian-Bashkirian seawater values (McArthur *et al.* 2010, 2012). The solution analyses show a spread in $^{87}\text{Sr}/^{86}\text{Sr}$ outside of the uncertainty of the individual measurements, indicating a slight compositional heterogeneity of the material. This heterogeneity is however well in the range of what is observed in other carbonate materials characterised for use as RM for Sr isotopic measurements (e.g., Weber *et al.* 2018, 2020).

The spatial homogeneity of Sr isotopes in SC-550 was tested by LA-MC-ICP-MS by performing forty static spot ablations across individual experimental charges using a small spot size ($38\ \mu\text{m}$). The three experiments run at $550\ ^\circ\text{C}$ (SC-C, SC-D, SC-E) display very good homogeneity in terms of $^{87}\text{Sr}/^{86}\text{Sr}$ ratios and mean values of individual experiments are within uncertainty of each other (Table 3, Figure 8a). Although the mean $^{84}\text{Sr}/^{86}\text{Sr}$ ratios obtained by LA-MC-ICP-MS are slightly lower than the natural ratio of 0.0565 and display scatter outside of the uncertainty of individual measurements, no systematic correlation is observed between the few anomalous $^{84}\text{Sr}/^{86}\text{Sr}$ ratios and the rare $^{87}\text{Sr}/^{86}\text{Sr}$ outliers (Figure 8b). The Rb/Sr ratio of the material is well below the threshold ($\text{Rb}/\text{Sr} = 0.03$) considered problematic for *in situ* LA-(MC)-ICP-MS measurements of Sr isotopes (Irrgeher *et al.* 2016, Mulder *et al.* 2023). Overall, SC-550 can be considered sufficiently homogeneous and suitable for *in situ* Sr isotope determination at the tens of μm scale.

The mean $^{87}\text{Sr}/^{86}\text{Sr}$ ratios obtained by LA-MC-ICP-MS on individual $550\ ^\circ\text{C}$ experimental charges are within uncertainty of the mean solution analyses of the starting nanoparticulate powder (Figure 8). As we observe no systematic offset between the Sr isotope ratio of starting nanoparticulate powder and experimental charges, we define the mean $^{87}\text{Sr}/^{86}\text{Sr}$ ratio of all solution-based measurements on the starting nanoparticulate powder of



0.7081041 ± 0.0000016 ($2s$, $n = 28$) as the information value for SC-550. However, to confirm the mean of solution analyses on nanoparticulate powder as the reference value for SC-550, further solution MC-ICP-MS analyses should be

carried out on an aliquot of sintered material (see Weber *et al.* 2020 for a similar approach), specifically considering the observed offset in Sr mass fraction between starting nanoparticulate powder and experimental charges (Figure 6).

Figure 9. Plots of U-Pb measurement results for IHPV experiments compared with original nanoparticulate powder plotted in Tera-Wasserburg concordia space. All $^{207}\text{Pb}/^{206}\text{Pb}$ ratios are normalised to NIST SRM 614 (Jochum *et al.* 2011), while $^{238}\text{U}/^{206}\text{Pb}$ ratios are normalised to NIST SRM 614 glass (Jochum *et al.* 2011) for original nanoparticulate pressed powder pellet and to WC-1 carbonate reference material (Roberts *et al.* 2017) for sintered carbonates. Grey ellipses represent nanoparticulate powder analyses, coloured ellipses represent IHPV experiment analyses. Panels (c), (d), (e) and (f) include group statistics. The large isotopic variability in experiments performed at higher temperature (i.e., SC-A, SC-B and to a lesser extent SC-G) is apparent. See Appendix S1 for further discussion on effect of Pb diffusion from the Ag capsule into sintered material at different experimental conditions. (g) Isotope dilution TIMS U-Pb results for five aliquots of original nanoparticulate powder, panel includes group statistics for ID-TIMS results.

Sintered calcite as RM for U-Pb geochronology

To date, no study has successfully employed synthetic (*sensu lato*) carbonate material as RM for monitoring and correction of U/Pb ratios in natural carbonates. The issues associated with using pressed carbonate powder pellets or doped amorphous carbonate include:

- (i) Heterogeneity in U and Pb mass fraction at the microscale (Ito *et al.* 2011), leading to heterogeneity in $^{238}\text{U}/^{206}\text{Pb}$ ratios (e.g., Miyajima *et al.* 2021), causing imprecision and/or inaccuracy in normalised U/Pb ratios;
- (ii) Differences in ablation efficiency between pressed power RM and unknown samples (e.g., Ito *et al.* 2011, Wu *et al.* 2018) leading to a mismatch in crater geometries, which affects the isotopic fractionation effects during ablation and leads to systematic inaccuracies in the normalised U/Pb ratios (Guillong *et al.* 2020).

In the following section, we discuss how our sintered carbonate material is characterised by homogeneity and ablation behaviour such that it can be used as matrix-matched RM for U/Pb ratio calibration in natural carbonates.

$^{207}\text{Pb}/^{206}\text{Pb}$ and $^{238}\text{U}/^{206}\text{Pb}$ – homogeneity and reference values

To test the spatial homogeneity in U/Pb ratios we performed ≥ 100 static spot ablations across individual 550 °C experimental charges with a large spot size routinely used for U-Pb dating of carbonates (110 μm ; Guillong *et al.* 2020). The analytical spots were randomised across the surface area and set along profiles to identify potential compositional gradients. In this section we only discuss

homogeneity in U/Pb isotopic ratios that are used for calculation of lower intercept ages in Tera-Wasserburg space, $^{207}\text{Pb}/^{206}\text{Pb}$ and $^{238}\text{U}/^{206}\text{Pb}$.

The results for randomly placed spots show very limited variations in $^{207}\text{Pb}/^{206}\text{Pb}$ ratios, ranging between 0.5% RSD and 0.6% RSD, and slightly larger variability in $^{238}\text{U}/^{206}\text{Pb}$ ratios, between 2.1% and 2.8%. Profiles across slices do not reveal any compositional gradients across the width of the experimental capsule (Figure S6). This observation is confirmed by U/Pb ratio mapping in SC-C, which reveals only a very narrow band (tens of μm) close to the outer edge characterised by a slightly lower $^{238}\text{U}/^{206}\text{Pb}$ ratio. This variation in U/Pb ratio is likely due to contamination during the experimental run by diffusion of Pb from the silver capsule to the sintered carbonate material (Figure 7c; see also Appendix S1 for further discussion). Overall, SC-550 can be considered homogenous for microanalytical U/Pb ratio determination at a scale of ca. 100 μm , which is considered optimal for LA-ICP-MS geochronology applications.

The mean $^{207}\text{Pb}/^{206}\text{Pb}$ ratios of SC-550 charges overlap within uncertainty and yield a mean of the means of 0.6644 ± 0.0019 (2s), while the mean of mean $^{238}\text{U}/^{206}\text{Pb}$ calculated for all experimental charges yields a value of 4.558 ± 0.054 (2s). These values are slightly offset from the $^{207}\text{Pb}/^{206}\text{Pb}$ (0.66226 ± 0.00016 , 2s) and $^{238}\text{U}/^{206}\text{Pb}$ (4.6244 ± 0.0057 , 2s) mean solution values, by 0.3% and 1.4%, respectively. This offset is well within the variability of individual measured ratios across experiments, where %RSD for $^{207}\text{Pb}/^{206}\text{Pb}$ ratios in SC-550 experiments varies from 0.5–0.6% and the %RSD for $^{238}\text{U}/^{206}\text{Pb}$ can be as large as 2.8% (e.g., SC-D). Given the agreement in U/Pb ratios obtained from solution and *in situ* techniques, we define the mean $^{207}\text{Pb}/^{206}\text{Pb}$ and $^{238}\text{U}/^{206}\text{Pb}$ ratio of all ID-TIMS measurements as the reference values for the SC-550 microanalytical material (Table 4). Reference values for

other U/Pb ratios (e.g., $^{235}\text{U}/^{207}\text{Pb}$ and $^{208}\text{Pb}/^{207}\text{Pb}$ that may be used in $^{207}\text{Pb}/^{235}\text{U}$ carbonate geochronology, Vermeesch *et al.* 2025) are included in Appendix S6.

Ablation efficiency and down-hole fractionation

The geometry of craters formed during ablation of carbonate materials affects the accuracy of U-Pb age determination (Guillong *et al.* 2020). Specifically, a mismatch in aspect ratio of ablation craters between RM and sample can lead to significant changes in U-Pb fractionation and consequent age offset. For identical ablation parameters (crater diameter, repetition rate, energy density), differences in ablation efficiency between RM and unknown sample are the main cause of mismatch in crater aspect ratio. We thus calculated ablation rates for our pressed powder pellet and all experimental charges. These ablation rates are then compared with the ablation rate of the natural calcite WC-1 (~120 nm per laser pulse at energy density of 2 J cm^{-2} ; Guillong *et al.* 2020), various carbonate materials (including calcite RMs) reported in Guillong *et al.* (2020) and other microanalytical reference materials compiled by Wu *et al.* (2018; Figure 10).

The pressed powder pellet produced in this study shows higher ablation efficiency than natural calcite material, varying from 110% to 130% relative to WC-1. The higher ablation rate in (nano-)powder pressed pellets (1.5 to 2.0 times higher than fusion glasses; Wu *et al.* 2018) and the general degraded ablation behaviour is likely caused by their lack of sintering, i.e., solely frictional adhesion of the powder particles. Conversely, the sintered carbonate products display lower ablation efficiency compared with natural calcite, with ablation rates that vary from 60% (SC-A) to 90% (SC-E) relative to WC-1 (Figure 10a). The range in ablation rates observed between experimental runs is likely a product of textural characteristics of the sintered material, in particular mean grain size, porosity and mean pore size (Figures 2 and 3; Wu *et al.* 2018). The ablation rates for the SC-550 charges are all within 15% of WC-1 and other natural calcite materials (Figure 10a). This indicates that at identical ablation conditions, there is a limited offset in crater aspect ratio between SC-550 and natural calcite samples (Figure 11a), which would correspond to a maximum offset on the lower intercept date of 1% or less if these experiments were used as RM for U/Pb ratio calibration (Guillong *et al.* 2020). Conversely, the higher temperature experiments, and especially the 800 °C series, yield lower ablation rates (i.e., 80% for SC-B and 60% for SC-A) that could result in substantial crater aspect ratio mismatch and consequent age offsets (i.e., 2 to 4%, Guillong *et al.* 2020) if this

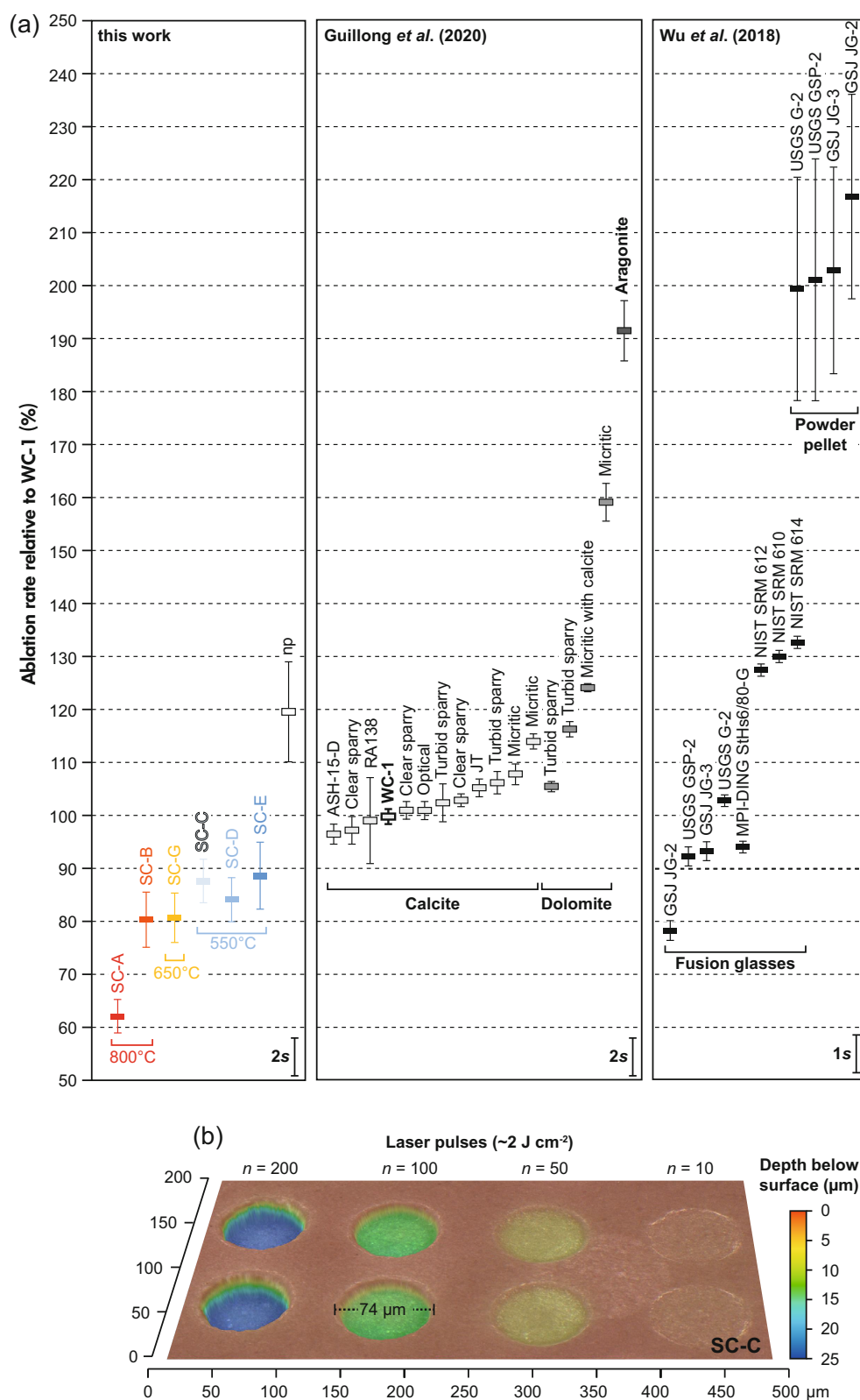
nanomaterial were used as primary RM for calibration of U/Pb isotopic ratios in natural calcite samples.

In addition to the ablation efficiency, we also tested the effect of down-hole fractionation (Nuriel *et al.* 2021) on U/Pb ratio during ablation of SC-550 (experiment SC-D) and compared it with a natural carbonate (WC-1) and a fusion glass (NIST SRM 614). The results of the test are shown both as (i) raw $^{206}\text{Pb}/^{238}\text{U}$ ratio changes vs. aspect ratio of the ablation crater (at variable ablation parameters; Figure 11a) and (ii) raw $^{206}\text{Pb}/^{238}\text{U}$ ratio variation vs. elapsed ablation time (at set ablation parameters, 51 μm spot and 20 Hz; Figure 11b). This comparison shows that down-hole fractionation of SC-550 is a closer match to WC-1 than it is to the NIST SRM 614 glass, even at “extreme” crater aspect ratios ≥ 1 (i.e., crater depth greater or equal to crater width).

Overall, these results highlight that SC-550 is functional for matrix-matched standardisation of U/Pb ratios in calcite, with both ablation rates and down-hole fractionation matching natural calcite materials more closely than other routinely used microanalytical reference materials (e.g., fused glasses of the NIST series). Ablation efficiency tests also reveal how SC-550 might be suboptimal as matrix-matched material for carbonates other than calcite (e.g., dolomite, aragonite), from which it exhibits large to very large offsets in ablation efficiency (Figure 10a). Therefore, using this sintered material as calibration RM outside of the calcite compositional/mineralogical range might result in the introduction of systematic inaccuracy (Guillong *et al.* 2020, Elisha *et al.* 2021).

Testing and validation

To test the suitability of SC-550 as a RM for U-Pb geochronology of natural carbonate materials, we analysed two calcite samples with known age using SC-550 (experiments SC-C and SC-E) as calibration RM. The U/Pb ratios of WC-1 (marine calcite cement; Roberts *et al.* 2017) and RA138 (microbial boundstone lithoclast, Guillong *et al.* 2024) were analysed with the same setup used to characterise the experimental products (see Method section and Appendix S1). Results are compiled over 30 analytical sessions during a one and a half year period (November 2024 – April 2026). The full dataset is provided in Appendix S8. Each session included twenty to fifty analyses of WC-1 and RA138. The raw $^{238}\text{U}/^{206}\text{Pb}$ and $^{207}\text{Pb}/^{206}\text{Pb}$ ratios were normalised based on the mean ratios measured on five aliquots of starting nanoparticulate powder by ID-TIMS (Table 4). Normalisation is performed in Iolite 4 (Paton *et al.* 2010, 2011) with the U-Pb



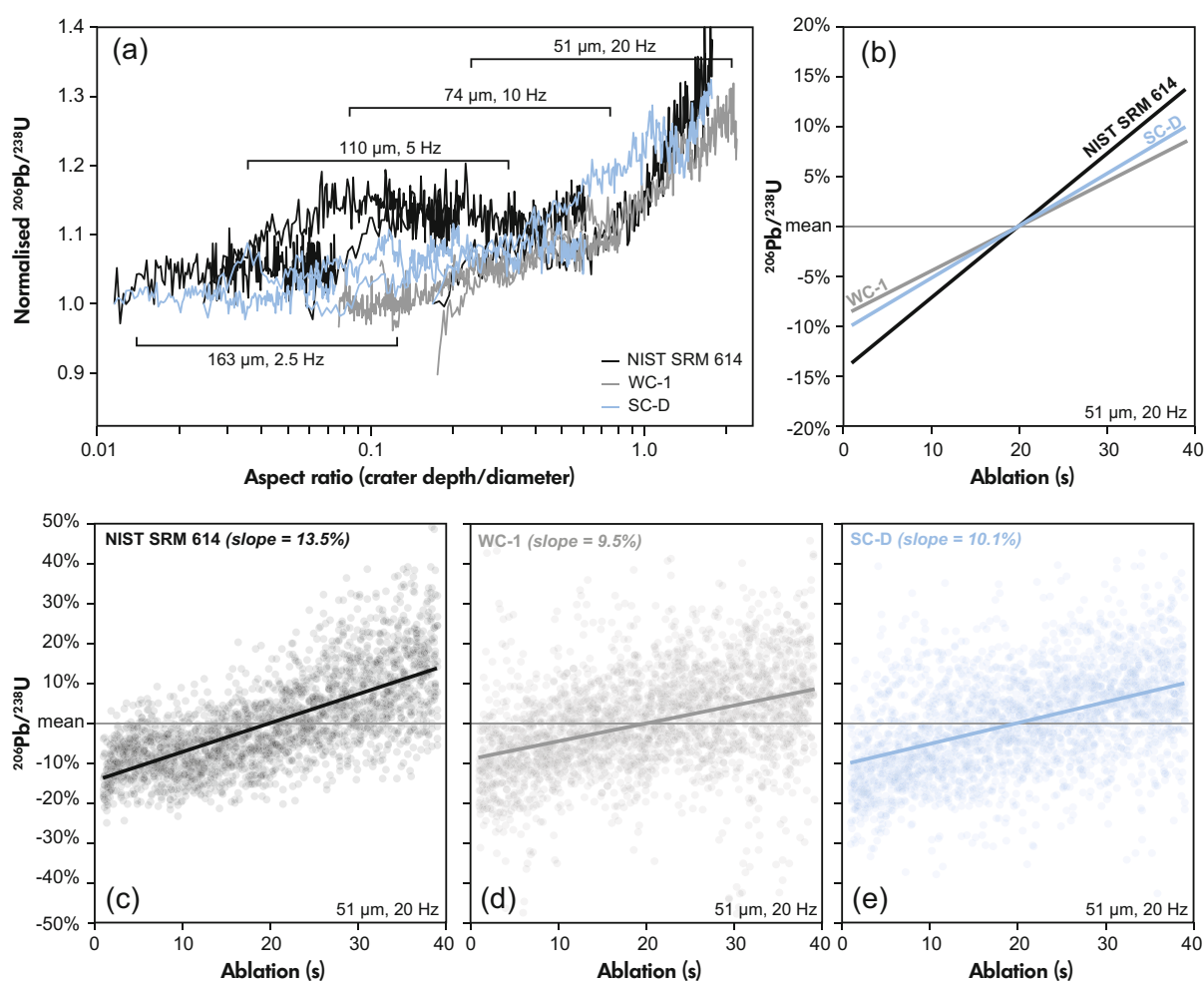


Figure 11. (a) Variation in normalised raw $^{206}\text{Pb}/^{238}\text{U}$ ratios with varying depth of the crater (expressed as crater aspect ratio) obtained for different analytical parameters on glass (NIST SRM 614), natural carbonate (WC-1) and IHPV sintered carbonate (SC-D). Panel (b) displays the difference in mean $^{206}\text{Pb}/^{238}\text{U}$ down-hole fractionation between (c) NIST SRM 614 (glass), (d) WC-1 (natural carbonate) and (e) one of the experimental runs at 550 °C (SC-D). Analyses were performed with a spot size of 51 μm , on sample fluence of 2 J cm^{-2} and 20 Hz repetition rate. Raw values (baseline corrected) are normalised to mean value, and a linear trend is fitted to show different fractionation trends between different materials used in the U-Pb data reduction process.

geochronology data reduction scheme and includes on-line baseline subtraction, drift correction and down-hole fractionation correction. Lower intercept dates were obtained by plotting the calibrated isotopic ratios on Tera-Wasserburg diagrams (IsoplotR, model 1 Discordia with 2s uncertainty, Vermeesch 2018). The $^{207}\text{Pb}/^{206}\text{Pb}$ intercept was anchored at 0.85 for WC-1 (Roberts *et al.* 2017) and 0.8459 for RA138 (Guillong *et al.* 2024).

The results for WC-1 and RA138 dating using SC-C and SC-E as primary RMs are summarised in Figure 12 and the results from individual sessions are included in

Appendix S8. For WC-1, the pooled Tera-Wasserburg isochron for $n = 702$ spot analyses (Figure 12a) yielded a lower intercept date of 254.9 ± 0.5 Ma (2s, not including propagation of excess uncertainty, S_{sys}) characterised by substantial inter- and intra-session dispersion (MSWD = 3.9). The thirty lower intercept dates calculated for individual sessions agree within uncertainty of the reference age, with dates ranging between 247.6 ± 3.2 | 5.9 Ma and 264.0 ± 2.4 | 5.8 Ma (2s, without | with inclusion of a $S_{\text{sys}} = 2\%$). The weighted mean age calculated from these thirty dates including 2% excess uncertainty is 254.1 ± 1.4 (MSWD = 1.7), within 0.1% of

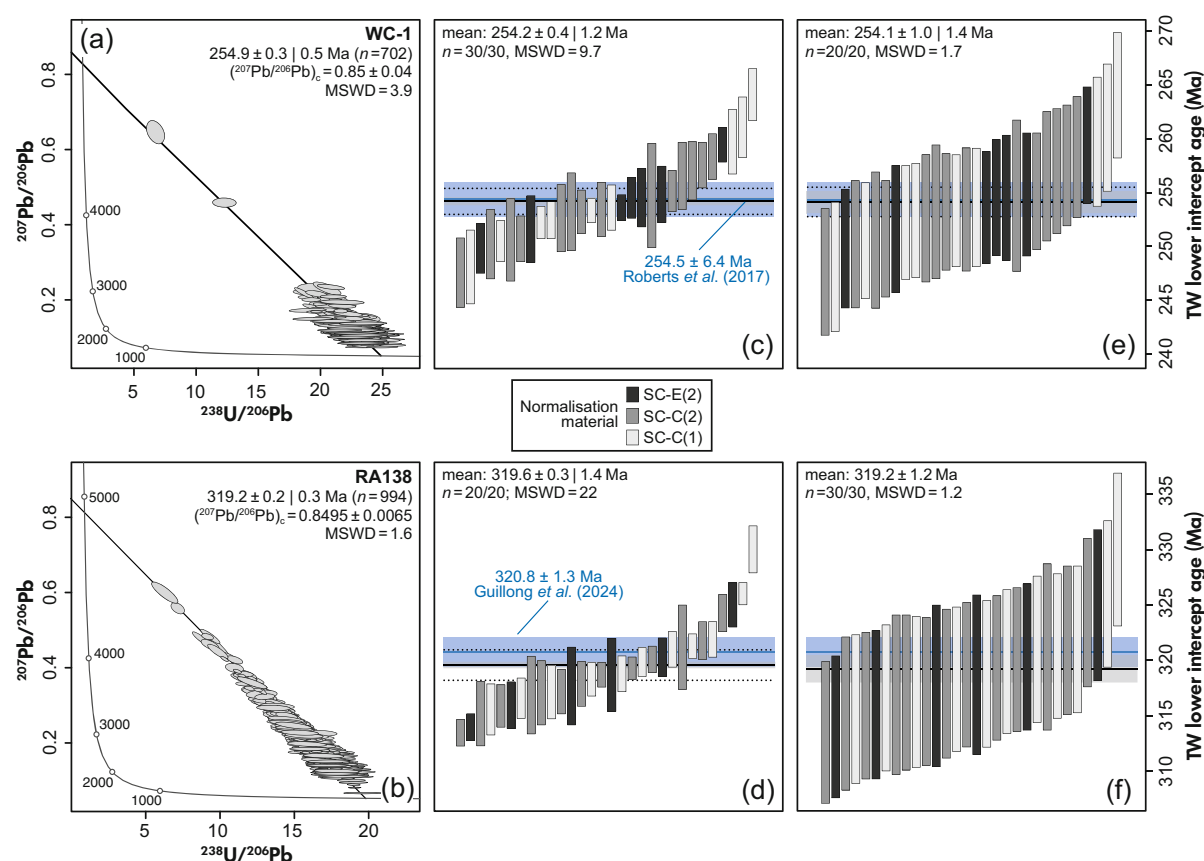


Figure 12. Validation of 550 °C sintering experimental charges (i.e., SC-C, SC-E) as primary RM for calibration of $^{207}\text{Pb}/^{206}\text{Pb}$ and $^{238}\text{U}/^{206}\text{Pb}$ ratios in U-Pb carbonate analyses. Left: lower intercept dates for (a) WC-1 and (b) RA138 on Tera-Wasserburg diagrams. Centre: rank order plot of lower intercept dates and weighted mean calculation for thirty sessions of (c) WC-1 and (d) RA138 showing excess scatter with MSWD of 10 and 22, respectively. Right: same data as in panels (c) and (d) but including a systematic uncertainty of 2%, resulting in a MSWD of 1.7 and 1.2 for (e) WC-1 and (f) RA138, respectively. The horizontal blue lines indicate the reference ages determined by ID-TIMS on WC-1 (254.4 ± 6.4 Ma, Roberts *et al.* 2017) and RA138 (320.8 ± 1.3 Ma, Guillong *et al.* 2024). Dotted lines and blue shaded areas denote 2s uncertainty of the results of this study and reference ages, respectively.

the published ID-TIMS age (Figure 12e; $254.4 \text{ Ma} \pm 6.4$, Roberts *et al.* 2017).

Calibration of RA138 using SC-550 as primary RM yielded a lower intercept age of 319.2 ± 0.3 Ma (MSWD = 1.6) for $n = 994$ spot analyses. The 30 individual lower intercept dates range between $313.5 \pm 1.2 \mid 6.4$ Ma and $330.0 \pm 2.1 \mid 6.9$ Ma (2s, without | with inclusion of a $S_{\text{sys}} = 2\%$), within 3% of the reference age. The weighted mean age of the thirty individual dates including S_{sys} is 319.2 ± 1.2 Ma (MSWD = 1.2), within uncertainty of the reference age (Figure 12e). The results obtained here suggest that the ID-TIMS discordia age for RA138 obtained in Guillong *et al.* (2024) by including all twenty-one sample aliquots (320.8 ± 1.3 Ma) might be more accurate than the

age that does not include four aliquots characterised by high common lead component (321.99 ± 0.65 Ma; see figure 4 of Guillong *et al.* 2024).

These results confirm the accuracy of using SC-550 as primary RM, as shown for an extended dataset for two independently calibrated calcite RMs. The results also show that SC-550 can be used to correct for LIEF as there are no obvious systematic down-hole effects in the final calculated U/Pb ratios. However, likely due to the slight heterogeneity of the sintered material in U and Pb (i.e., 2–3% RSD for $^{238}\text{U}/^{206}\text{Pb}$ ratio, Figure 9 and Table 4) it is necessary to add 2% excess uncertainty (S_{sys}) to the individual lower intercept dates in order to obtain weighted mean age not characterised by over dispersion (Figure 12). This amount of

excess uncertainty is characteristic for both carbonate (Guillong *et al.* 2024) and zircon LA-ICP-MS data (Sliwinski *et al.* 2022) where long-term compilations are available.

Recommended use and availability of SC-550

The use of SC-550 as primary RM for U-Pb geochronology simplifies the data reduction procedure. The two-step normalisation method employed until now (Roberts *et al.* 2017) includes: (i) the correction of $^{207}\text{Pb}/^{206}\text{Pb}$ using NIST glass followed by (ii) the correction of $^{238}\text{U}/^{206}\text{Pb}$ by multiplying the $^{238}\text{U}/^{206}\text{Pb}$ of each analysis by a session-specific factor, which is based on the comparison of the measured and accepted lower intercept age of a carbonate with known age (e.g., WC-1, Roberts *et al.* 2017). With SC-550, this procedure is reduced to a single down-hole fractionation and normalisation step for all ratios based on a homogenous, matrix-matched carbonate material. This can be done in most data-reduction software (e.g., U-Pb geochronology data reduction scheme in Lolite 4, Paton *et al.* 2010, 2011). Moreover, the high U and Pb content ($1\text{--}4\text{ }\mu\text{g g}^{-1}$) of SC-550 makes it a viable RM for LA-ICP-MS systems with lower sensitivities (e.g., Quadrupole-ICP-MS), compared with other routinely used carbonate RMs (e.g., ASH-15D, Nuriel *et al.* 2021). We recommend using a large spot size ($> 100\text{ }\mu\text{m}$) to minimise down-hole fractionation effects (Guillong *et al.* 2020) and avoid potential issues with matrix heterogeneity at the smaller scale (tens of μm). Individual time-resolved signals should also be screened for occasional anomalous Pb-rich or U-rich domains whose inclusion in the selection interval could lead to larger uncertainty in the calculated U/Pb isotopic ratios and therefore propagated to the final age of the unknown sample (i.e., before excess uncertainty propagation). Finally, we observed elevated Hg intensity in a sample mounted with the Ag capsule; therefore we recommend mounting SC-550 without the silver capsule.

In total five experiments were successfully conducted at optimal sintering conditions (i.e., 500 MPa, 550 °C, 24 h, Table 1), three of which have been fully characterised as shown above. Each experimental charge resulted in a sintered carbonate volume of ca. 2.5 cm^3 , which was cut in eighteen to twenty individual slices ca. 2 mm in thickness. This material is intended to be widely used as a reference material for *in situ* measurements of U-Pb and potentially as quality control for Sr isotope and trace element measurements. Individual SC-550 slices are available on request from Lorenzo Tavazzani or Marcel Guillong. Further, with ca. 220 g of starting nanoparticulate powder left, a sufficient

amount of material is available for further syntheses of SC-550.

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Data availability statement

The authors confirm that all data supporting the findings of this study are available within the article and its supplementary materials.

References

- Addadi L, Raz S. and Weiner S. (2003) Taking advantage of disorder: Amorphous calcium carbonate and its roles in biomineralisation. *Advanced Materials*, 15, 959–970.
- Aguirre Palafox L.E., Möller A., McLean N.M., Ludvigson G.A., Colombi C.E. and Montañez I.P. (2024) U-Pb geochronology of paleosol carbonate cements by LA-ICP-MS: A proof of concept and strategy for dating the terrestrial record. *Geochemistry, Geophysics, Geosystems*, 25, e2024GC011488.
- Berndt J., Liebske C., Holtz F., Freise M., Nowak M., Ziegenbein D., Hurkuck W. and Koepke J. (2002) A combined rapid-quench and H₂-membrane setup for internally heated pressure vessels: Description and application for water solubility in basaltic melts. *American Mineralogist*, 87, 1717–1726.
- Boer W., Nordstad S., Weber M., Mertz-Kraus R., Hönisch B., Bijma J., Raitzsch M., Wilhelms-Dick D., Foster G.L., Goring-Harford H., Nürnberg D., Hauff F., Kuhnert H., Lugli F., Spero H., Rosner M., Van Gaever P., De Nooijer L.J. and Reichart G. (2022) New calcium carbonate nano-particulate pressed powder pellet (NFHS-2-NP) for LA-ICP-OES, LA-(MC)-ICP-MS and μXRF . *Geostandards and Geoanalytical Research*, 46, 411–432.

references

- Bos Orent E., Barton M.D. and Kirk J.D. (2025)**
Episodic copper mineralization in sedimentary basins: Evidence from the Paradox Basin, USA. *Earth and Planetary Science Letters*, 671, 119620.
- Burisch M., Gerdes A., Walter B.F., Neumann U., Fettel M. and Markl G. (2017)**
Methane and the origin of five-element veins: Mineralogy, age, fluid inclusion chemistry and ore forming processes in the Odenwald, SW Germany. *Ore Geology Reviews*, 81, 42–61.
- Byerly G.R., Lower D.R. and Walsh M.M. (1986)**
Stromatolites from the 3,300–3,500-Myr Swaziland Supergroup, Barberton Mountain Land, South Africa. *Nature*, 319, 489–491.
- Chiaradia M., Müntener O. and Beate B. (2020)**
Effects of aseismic ridge subduction on the geochemistry of frontal arc magmas. *Earth and Planetary Science Letters*, 531, 115984.
- Condon D.J., Schoene B., McLean N.M., Bowring S.A. and Parrish R.R. (2015)**
Metrology and traceability of U–Pb isotope dilution geochronology (EARTHTIME Tracer Calibration Part I). *Geochimica et Cosmochimica Acta*, 164, 464–480.
- Covey-Crump S.J. (1997)**
The normal grain growth behaviour of nominally pure calcitic aggregates. *Contributions to Mineralogy and Petrology*, 129, 239–254.
- Davis N.E., Newman J., Wheelock P.B. and Kronenberg A.K. (2011)**
Grain growth kinetics of dolomite, magnesite and calcite: A comparative study. *Physics and Chemistry of Minerals*, 38, 123–138.
- Drake H., Heim C., Roberts N.M.W., Zack T., Tillberg M., Broman C., Ivarsson M., Whitehouse M.J. and Åström M.E. (2017)**
Isotopic evidence for microbial production and consumption of methane in the upper continental crust throughout the Phanerozoic eon. *Earth and Planetary Science Letters*, 470, 108–118.
- Elisha B., Nuriel P., Kylander-Clark A. and Weinberger R. (2021)**
Towards *in situ* U–Pb dating of dolomite. *Geochronology*, 3, 337–349.
- Fitzpayne A., Giuliani A., Howarth G.H., Peters B.J., Fehr M.A. and Maas R. (2023)**
Major, trace-element and Sr–Nd–Hf isotope geochemistry of diamondiferous dykes from Tonguma and Koidu, Sierra Leone: Highly micaceous kimberlites formed by assimilation of metasomatised lithospheric mantle rocks. *Chemical Geology*, 630, 121475.
- Garbe-Schönberg D. and Müller S. (2014)**
Nano-particulate pressed powder tablets for LA-ICP-MS. *Journal of Analytical Atomic Spectrometry*, 29, 990–1000.
- Giomo M., Barale L., Bertok C., Frenzel M., Looser N., Guillong M., Bemasoni S.M. and Martire L. (2022)**
Sulfide-associated hydrothermal dolomite and calcite reveal a shallow burial depth for Alpine-type Zn–(Pb) deposits. *Geology*, 50, 853–858.
- Godeau N., Deschamps P., Guihou A., Leonide P., Tendil A., Gerdes A., Hamelin B. and Girard J.-P. (2018)**
U–Pb dating of calcite cement and diagenetic history in microporous carbonate reservoirs: Case of the Urganian Limestone, France. *Geology*, 46, 247–250.
- Guillong M., Samankassou E., Müller I.A., Szymanowski D., Looser N., Tavazzani L., Merino-Tomé Ó., Bahamonde J.R., Buret Y. and Ovtcharova M. (2024)**
Technical note: RA138 Calcite U–Pb LA-ICP-MS primary reference material. *Geochronology*, 6, 465–474.
- Guillong M., Wotzlaw J.-F., Looser N. and Laurent O. (2020)**
Evaluating the reliability of U–Pb laser ablation inductively coupled plasma-mass spectrometry (LA-ICP-MS) carbonate geochronology: Matrix issues and a potential calcite validation reference material. *Geochronology*, 2, 155–167.
- Hansman R.J., Albert R., Gerdes A. and Ring U. (2018)**
Absolute ages of multiple generations of brittle structures by U–Pb dating of calcite. *Geology*, 46, 207–210.
- Hathorne E.C., James R.H., Savage P. and Alard O. (2008)**
Physical and chemical characteristics of particles produced by laser ablation of biogenic calcium carbonate. *Journal of Analytical Atomic Spectrometry*, 23, 240–243.
- Hielscher R. and Schaebe H. (2008)**
A novel pole figure inversion method: specification of the MTEX algorithm. *Journal of Applied Crystallography*, 41, 1024–1037.
- Horstwood M.S.A., Košler J., Gehrels G., Jackson S.E., McLean N.M., Paton C., Pearson N.J., Sircombe K., Sylvester P., Vermeesch P., Bowring J.F., Condon D.J. and Schoene B. (2016)**
Community-derived standards for LA-ICP-MS U–(Th)–Pb geochronology – Uncertainty propagation, age interpretation and data reporting. *Geostandards and Geoanalytical Research*, 40, 311–332.
- Horwitz W. and Albert R. (1995)**
Precision in analytical measurements: Expected values and consequences in geochemical analyses. *Fresenius' Journal of Analytical Chemistry*, 351, 507–513.
- Irgeher J., Galler P. and Prohaska T. (2016)**
⁸⁷Sr/⁸⁶Sr isotope ratio measurements by laser ablation multicollector inductively coupled plasma-mass spectrometry: Reconsidering matrix interferences in bioapatites and biogenic carbonates. *Spectrochimica Acta Part B*, 125, 31–42.



references

- Ito K., Hasebe N., Hasebe A. and Arai S. (2011)
The matrix effect on ^{238}U and ^{232}Th measurements using pressed powder pellets by LA-ICP-MS. *Geochemical journal*, 45, 375–385.
- Jahn B. and Cuvelier H. (1994)
Pb-Pb and U-Pb geochronology of carbonate rocks: An assessment. *Chemical Geology*, 115, 125–151.
- Jochum K.P. and Enzweiler J. (2014)
Reference materials in geochemical and environmental research. In: Turekian K.K. (ed.), *Treatise on geochemistry* (2nd edition). Elsevier (Oxford), 43–70.
- Jochum K.P., Garbe-Schönberg D., Vetter M., Stoll B., Weis U., Weber M., Lugli F., Jentzen A., Schiebel R., Wassenburg J.A., Jacob D.E. and Haug G.H. (2019)
Nano-powdered calcium carbonate reference materials: Significant progress for microanalysis? *Geostandards and Geoanalytical Research*, 43, 595–609.
- Jochum K.P., Scholz D., Stoll B., Weis U., Wilson S.A., Yang Q., Schwalb A., Bomer N., Jacob D.E. and Andreae M.O. (2012)
Accurate trace element analysis of speleothems and biogenic calcium carbonates by LA-ICP-MS. *Chemical Geology*, 318–319, 31–44.
- Jochum K.P., Stoll B., Weis U., Jacob D.E., Mertz-Kraus R. and Andreae M.O. (2014)
Non-matrix-matched calibration for the multi-element analysis of geological and environmental samples using 200 nm femtosecond LA-ICP-MS: A comparison with nanosecond lasers. *Geostandards and Geoanalytical Research*, 38, 265–292.
- Jochum K.P., Weis U., Stoll B., Kuzmin D., Yang Q., Raczek I., Jacob D.E., Stracke A., Birbaum K. and Frick D.A. (2011)
Determination of reference values for NIST SRM 610–617 glasses following ISO guidelines. *Geostandards and Geoanalytical Research*, 35, 397–429.
- Koga N., Nakagoe Y. and Tanaka H. (1998)
Crystallization of amorphous calcium carbonate. *Thermochimica Acta*, 318, 239–244.
- Li Q., Parrish R.R., Horstwood M.S.A. and McArthur J.M. (2014)
U–Pb dating of cements in Mesozoic ammonites. *Chemical Geology*, 376, 76–83.
- Li T., Chen R., Wu S., Romer R.L., Roberts N.M.W., Drost K., Chew D., Guillong M., Ma Q., Yue Y., Li S., Xie B., Abdurahman A., Wang H., Lin H., Yang Y. and Guo X. (2026)
Calcite TLM as a new natural reference material for *in situ* U–Pb geochronology and its geological implications. *Chemical Geology*, 703, 123230.
- Looser N., Aschwanden L., Wohlwend S., Guillong M., Rickli J., Schneeberger R. and Bernasconi S.M. (2025)
Constraining 200 million years of geodynamic evolution of the North Alpine foreland at million-year resolution using clumped isotopes and U–Pb dating of diagenetic carbonates. *Earth and Planetary Science Letters*, 667, 119410.
- Looser N., Madritsch H., Guillong M., Laurent O., Wohlwend S. and Bernasconi S.M. (2021)
Absolute age and temperature constraints on deformation along the basal décollement of the Jura fold-and-thrust belt from carbonate U–Pb dating and clumped isotopes. *Tectonics*, 40, e2020TC006439.
- Lowe D.R. (1980)
Stromatolites 3,400-Myr old from the Archean of Western Australia. *Nature*, 284, 441–443.
- Markowska M., Vonhof H.B., Groucutt H.S., Breeze P.S., Drake N., Stewart M., Albert R., Andrieux E., Blinkhorn J., Boivin N., Budsky A., Clark-Wilson R., Fleitmann D., Gerdes A., Martin A.N., Martínez-García A., Nicholson S.L., Price G.J., Scerri E.M.L., Scholz D., Vanwezer N., Weber M., Alsharekh A.M., Al Omari A.A., Al-Mufarreh Y.S.A., Al-Jibreen F., Alqahtani M., Al-Shanti M., Zalmout I., Petraglia M.D. and Haug G.H. (2025)
Recurrent humid phases in Arabia over the past 8 million years. *Nature*, 640, 954–961.
- McArthur J., Howarth R.J. and Shields G.A. (2012)
Strontium isotope stratigraphy. In: Gradstein F.M., Ogg J.G., Schmitz M. and Ogg G. (eds), *The Geological Time Scale*. Elsevier, 127–144.
- McArthur J.M. (2010)
Strontium isotope stratigraphy. Application of Modern Stratigraphic Techniques: Theory and Case Histories, 94.
- McLean N.M., Condon D.J., Schoene B. and Bowring S.A. (2015)
Evaluating uncertainties in the calibration of isotopic reference materials and multi-element isotopic tracers (EARTHTIME Tracer Calibration Part II). *Geochimica et Cosmochimica Acta*, 164, 481–501.
- Meece D.E. and Benninger L.K. (1993)
The coprecipitation of Pu and other radionuclides with CaCO_3 . *Geochimica et Cosmochimica Acta*, 57, 1447–1458.
- Methner K., Mulch A., Fiebig J., Wacker U., Gerdes A., Graham S.A. and Chamberlain C.P. (2016)
Rapid Middle Eocene temperature change in western North America. *Earth and Planetary Science Letters*, 450, 132–139.
- Miyajima Y., Saito A., Kagi H., Yokoyama T., Takahashi Y. and Hirata T. (2021)
Incorporation of U, Pb and rare earth elements in calcite through crystallisation from amorphous calcium carbonate: Simple preparation of reference materials for microanalysis. *Geostandards and Geoanalytical Research*, 45, 189–205.
- Montano D., Gasparini M., Gerdes A., Della Porta G. and Albert R. (2021)
In-situ U–Pb dating of Ries Crater lacustrine carbonates (Miocene, south-west Germany): Implications for continental carbonate chronostratigraphy. *Earth and Planetary Science Letters*, 568, 117011.

references

- Mottram C.M., Kellett D.A., Barresi T., Chapman G.G. and Halle J. (2024)
Tracking the porphyry-epithermal mineralization transition using U-Pb carbonate dating. *Geology*, 52, 723–728.
- Mulder J., Hagen-Peter G., Ubide T., Andreassen R., Kooijman E., Kielman-Schmitt M., Feng Y., Paul B., Karlsson A., Tegner C., Leshner C. and Costa F. (2023)
New reference materials, analytical procedures and data reduction strategies for Sr isotope measurements in geological materials by LA-MC-ICP-MS. *Geostandards and Geoanalytical Research*, 47, 311–336.
- Norris C.A., Danyushevsky L., Olin P. and West N.R. (2021)
Elimination of aliasing in LA-ICP-MS by alignment of laser and mass spectrometer. *Journal of Analytical Atomic Spectrometry*, 36, 733–739.
- Nuriel P., Craddock J., Kylander-Clark A.R.C., Uysal I.T., Karabacak V., Dirik R.K., Hacker B.R. and Weinberger R. (2019)
Reactivation history of the North Anatolian fault zone based on calcite age-strain analyses. *Geology*, 47, 465–469.
- Nuriel P., Weinberger R., Kylander-Clark A.R.C., Hacker B.R. and Craddock J.P. (2017)
The onset of the Dead Sea transform based on calcite age-strain analyses. *Geology*, 45, 587–590.
- Nuriel P., Wotzlaw J.-F., Ovtcharova M., Vaks A., Stremtan C., Sala M., Roberts N.M.W. and Kylander-Clark A.R.C. (2021)
The use of ASH-15 flowstone as a matrix-matched reference material for laser-ablation U-Pb geochronology of calcite. *Geochronology*, 3, 35–47.
- Olgaard D.L. and Evans B. (1988)
Grain growth in synthetic marbles with added mica and water. *Contributions to Mineralogy and Petrology*, 100, 246–260.
- Paton C., Hellstrom J., Paul B., Woodhead J. and Hergt J. (2011)
Iolite: Freeware for the visualisation and processing of mass spectrometric data. *Journal of Analytical Atomic Spectrometry*, 26, 2508.
- Paton C., Woodhead J.D., Hellstrom J.C., Hergt J.M., Greig A. and Maas R. (2010)
Improved laser ablation U-Pb zircon geochronology through robust downhole fractionation correction. *Geochemistry, Geophysics, Geosystems*, 11, Q0AA06.
- Qi L., Hou M., Mulder J.A., Cawood P.A., Guo Y., Wu S., Jiao L., Zhang X. and Ouyang H. (2024)
In situ U-Pb dating of Jurassic dinosaur bones from Sichuan Basin, south China. *Geology*, 52, 216–221.
- Rasbury E.T. and Cole J.M. (2009)
Directly dating geologic events: U-Pb dating of carbonates. *Reviews of Geophysics*, 47, RG3001.
- Reeder R.J., Nugent M., Lamble G.M., Tait C.D. and Morris D.E. (2000)
Uranyl incorporation into calcite and aragonite: XAFS and luminescence studies. *Environmental Science and Technology*, 34, 638–644.
- Reeder R.J., Nugent M., Tait C.D., Morris D.E., Heald S.M., Beck K.M., Hess W.P. and Lanzietti A. (2001)
Coprecipitation of uranium(VI) with calcite: XAFS, microXAS, and luminescence characterization. *Geochimica et Cosmochimica Acta*, 65, 3491–3503.
- Roberts N.M.W., Drost K., Horstwood M.S.A., Condon D.J., Chew D., Drake H., Milodowski A.E., McLean N.M., Smye A.J., Walker R.J., Haslam R., Hodson K., Imber J., Beaudoin N. and Lee J.K. (2020)
Laser ablation inductively coupled plasma-mass spectrometry (LA-ICP-MS) U-Pb carbonate geochronology: Strategies, progress, and limitations. *Geochronology*, 2, 33–61.
- Roberts N.M.W., Rasbury E.T., Parrish R.R., Smith C.J., Horstwood M.S.A. and Condon D.J. (2017)
A calcite reference material for LA-ICP-MS U-Pb geochronology. *Geochemistry Geophysics Geosystems*, 18, 2807–2814.
- Roberts N.M.W. and Walker R.J. (2016)
U-Pb geochronology of calcite-mineralized faults: Absolute timing of rift-related fault events on the northeast Atlantic margin. *Geology*, 44, 531–534.
- Scardia G., Parenti F., Miggins D.P., Gerdes A., Araujo A.G.M. and Neves W.A. (2019)
Chronologic constraints on hominin dispersal outside Africa since 2.48 Ma from the Zarqa Valley Jordan. *Quaternary Science Reviews*, 219, 1–19.
- Sliwinski J.T., Guillong M., Horstwood M.S.A., Bachmann O. (2022)
Quantifying long-term reproducibility of zircon reference materials by U-Pb LA-ICP-MS dating. *Geostandards and Geoanalytical Research*, 46, 401–409.
- Sylvester P.J. (2008)
Matrix effects in laser ablation-ICP-MS. In: Sylvester P.J. (ed.), *Laser ablation ICP-MS in the Earth sciences: Current practices and outstanding issues*. Mineralogical Association of Canada, Short Course Series, 40, 67–78.
- Tabersky D., Luechinger N.A., Rossier M., Reusser E., Hametner K., Aeschlimann B., Frick D.A., Halim S.C., Thompson J. and Danyushevsky L. (2014)
Development and characterization of custom-engineered and compacted nanoparticles as calibration materials for quantification using LA-ICP-MS. *Journal of Analytical Atomic Spectrometry*, 29, 955–962.
- Tavazzani L., Guillong M., Giuliani A., Fontboté L. and Chelle-Michou C. (2024)
Not so fast: Million-years of metal precipitation in Mississippi Valley type deposits inferred from *in-situ* petrochronology of hydrothermal carbonates. *Earth and Planetary Science Letters*, 636, 118718.

references

- Vermeesch P. (2018)**
IsoplotR: A free and open toolbox for geochronology. *Geoscience Frontiers*, 9, 1479–1493.
- Vermeesch P., McLean N., Vaks A., Golan T., Breitenbach S.F.M. and Parrish R. (2025)**
Carbonate $^{206}\text{Pb}/^{238}\text{U}$ problems and potential $^{207}\text{Pb}/^{235}\text{U}$ fixes. *Geochronology*, 7, 459–473.
- Weber M., Lugli F., Hattendorf B., Scholz D., Mertz-Kraus R., Guinoiseau D. and Jochum K.P. (2020)**
NanoSr – A new carbonate microanalytical reference material for *in situ* strontium isotope analysis. *Geostandards and Geoanalytical Research*, 44, 69–83.
- Weber M., Lugli F., Jochum K.P., Cipriani A. and Scholz D. (2018)**
Calcium carbonate and phosphate reference materials for monitoring bulk and microanalytical determination of Sr isotopes. *Geostandards and Geoanalytical Research*, 42, 77–89.
- Wu S., Karius V., Schmidt B.C., Simon K. and Wörner G. (2018)**
Comparison of ultrafine powder pellet and flux-free fusion glass for bulk analysis of granitoids by laser ablation-inductively coupled plasma-mass spectrometry. *Geostandards and Geoanalytical Research*, 42, 575–591.
- Yoshino T., Maruyama K., Kagi H., Nara M. and Kim J.C. (2012)**
Pressure-induced crystallization from amorphous calcium carbonate. *Crystal Growth and Design*, 12, 3357–3361.

Supporting information

The following supporting information may be found in the online version of this article:

Appendix S1. Supplementary text and figures, including: Extended analytical methods, trace element data set evaluations and description of preliminary, small-volume experiments.

Figure S1. Characterization of the different phases found in RA138.

Figure S2. Top view, optical microscope image of slices from the central section of three experimental charges.

Figure S3. Trace element distribution maps (Sr, Fe, Mn, Mg, Pb, U) for three experimental charges.

Figure S4. Variability in trace elements on individual experimental charges characterised in this study.

Figure S5. LA-ICP-MS U-Pb results for IHPV experiments compared with original nanoparticulate powder plotted in Tera-Wasserburg concordia space.

Figure S6. Results of analytical profiles across slices of three different experimental charges.

Figure S7. BSE images of piston cylinder apparatus (PCA) experiments performed at 800 °C, 1 GPa and 1 h, and 800 °C, 2 GPa and 24 h.

Figure S8. LA-ICP-MS U-Pb results for three PCA experiments.

Appendix S2. EPMA data (ETH-Zürich).

Appendix S3. LA-ICP-MS trace element mass fraction data (ETH-Zürich).

Appendix S4. MC-ICP-MS Sr isotopic data (Geneva).

Appendix S5. LA-MC-ICP-MS Sr isotopic data (ETH-Zürich).

Appendix S6. ID-TIMS U-Pb data (ETH-Zürich).

Appendix S7. LA-ICP-MS U-Pb data (ETH-Zürich).

Appendix S8. LA-ICP-MS U-Pb data – Validation data set (ETH-Zürich).

Appendix S9. Extended experimental conditions table.

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